

EV POWERTRAIN

Cooling & Lubrication System

Systems Engineering Reference

Complete Reference Guide

- System Architecture & Cooling Circuits
- Oil Pump Sizing & Flow Rate Engineering
- Thermal Model (LPTN) & Derating Calibration
- Drive Cycles & V-Model Verification
- Requirements Engineering & EARS Syntax
- Functional Safety, HARA & DFMEA
- Oil Lifetime Profile & Arrhenius Analysis
- Winding & Magnet Temperature Management

CHAPTER

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PART I

Simulation Toolchains

CarSim, CarMaker & Related Tools

CHAPTER 1

CarSim — Vehicle Dynamics Simulation

Mechanical Simulation Corporation | Industry-Standard Platform

1.1 What is CarSim?

CarSim is a commercial vehicle dynamics simulation software developed and maintained by Mechanical Simulation Corporation (MSC). It delivers highly accurate, detailed, and computationally efficient methods for simulating the performance of passenger vehicles, light-duty trucks, and sports cars. The tool is used throughout the entire vehicle development lifecycle — from initial concept exploration through to hardware sign-off and production release.

The tool has been in continuous development since its first commercial release in the early 1990s. Its mathematical models are based on decades of university research in vehicle dynamics and have been extensively validated against physical test data from OEM development programmes worldwide.

Note: CarSim is used extensively by 7 of the 10 largest automotive OEMs globally.

1.2 Key Capabilities

- Accurate simulation of braking, handling, ride quality, vehicle stability and acceleration
- Over 15 built-in vehicle types with approximately 80 parametric variants
- More than 600 example simulation configurations covering a wide range of manoeuvres
- Used by over 110 OEMs, Tier 1 suppliers, 200+ universities and government research labs
- Standalone operation: no third-party solver required to run simulations
- Economical licensing compared to other commercial vehicle dynamics packages
- VS Visualizer tool for photo-realistic 3D animation and post-processing plots
- VS Commands scripting language for run-time user-defined logic
- Hardware-in-the-Loop (HIL) support on major real-time platforms
- Driving simulator integration — engineers feel subtle performance attributes

1.3 Integration Interfaces

CarSim is designed as an open integration platform. Engineers connect it to their preferred modelling and control environments through standardised interfaces:

- MATLAB/Simulink — industry-standard control and algorithm development platform
- NI LabVIEW — National Instruments data acquisition and test automation platform
- FMI/FMU (Functional Mockup Interface / Unit) — open standard for model exchange; CarSim can automatically generate FMUs for use in any FMI-compliant environment
- VS SDK (Software Development Kit) — allows custom programmes in MATLAB, C/C++, Visual Basic to be linked to the CarSim solver
- Windows COM and remote control interfaces for automated batch simulation
- Real-time platforms for HIL test benches

1.4 Simulation Features

CarSim simulates the full vehicle as a parametric multi-body system. The core simulation computes tyre forces, suspension kinematics and compliance, steering geometry, powertrain response, brake distribution, and aerodynamic loads at every timestep. Built-in driver models provide:

- Path-following controller — follows a target road trajectory
- Speed and acceleration control — maintains target velocity profiles
- Gear shifting and mechanical clutch control — for manual and automatic transmissions
- Complex scenario definition — multi-event sequences on configurable road surfaces
- Statistical and sensitivity studies — parameter sweeps for design of experiments

Note: For eDU powertrain development, CarSim is commonly used to generate representative torque-demand and speed profiles that become inputs to eDU thermal and efficiency models.

CHAPTER 2

CarMaker — Open Integration & Test Platform

IPG Automotive GmbH | Full-System Co-Simulation

2.1 What is CarMaker?

CarMaker is a vehicle simulation and test platform developed by IPG Automotive GmbH, headquartered in Karlsruhe, Germany. Unlike CarSim, which focuses primarily on vehicle dynamics, CarMaker is architected as an open integration platform — designed to act as the simulation backbone into which subsystem models, sensor models, controller software, and driver models can all be integrated and co-simulated.

The CarMaker product family includes CarMaker (passenger vehicles), TruckMaker (heavy commercial vehicles), and MotorcycleMaker (two-wheelers). CarMaker 15.0 (2026) is the current release, featuring enhanced ground truth sensors, a new battery model, and modular powertrain architecture for BEV

development.

2.2 Core Architecture

- IPGDriver — proprietary driver model simulating reaction times, personal judgement, emotional state, anticipation, and fatigue
- Scenario Editor — graphical tool for building detailed road networks and traffic scenarios including vegetation, buildings, traffic signs, terrain, and weather
- Sensor Models — camera, lidar, radar, ultrasonic, and ground truth sensor models for ADAS and autonomous system development
- IPGControl — data analysis and graphical post-processing environment
- MovieNX — realistic 3D animation and rendering with Python API for automation
- FMI/FMU Interface — import and export of models in the open FMI standard
- MATLAB/Simulink Integration — tight co-simulation via S-function blocks
- AWS and Cloud Integration — for large-scale parallel simulation campaigns

2.3 Data Generation in CarMaker

One of CarMaker's growing strengths is its use as a synthetic data generation platform for training and validating machine learning and AI-based functions. CarMaker can produce large volumes of labelled, annotated simulation data that would be prohibitively expensive or physically impossible to capture in the real world.

Ground Truth Sensor

CarMaker 12 introduced a dedicated ground truth sensor with multiple data access modes. This sensor provides pixel-perfect labels for object detection, lane detection, depth estimation, and semantic segmentation directly from the simulation environment.

Simulation Output Channels

- Vehicle state: position, velocity, acceleration, yaw/pitch/roll rates
- Powertrain: torque, speed, gear state, regenerative braking power
- Thermal: motor temperature, inverter temperature, battery SOC and temperature
- Sensor outputs: camera images, lidar point clouds, radar detections
- Ground truth: object bounding boxes, lane boundaries, depth maps, semantic labels
- Road and environment: friction coefficients, gradient, curvature, traffic state

2.4 CarMaker vs CarSim Comparison

Attribute	CarSim	CarMaker
Developer	Mechanical Simulation Corp.	IPG Automotive GmbH
Primary Focus	Vehicle dynamics accuracy	Open integration & full-system test
ADAS/AD Development	Limited	Extensive (sensor models, scenarios)
Synthetic Data Generation	Not primary use case	Core capability
Driver Model	Built-in path/speed controllers	IPGDriver (human behaviour model)

Attribute	CarSim	CarMaker
Scenario Editor	Basic	Advanced (3D, traffic, weather, terrain)
Cloud Scalability	Limited	AWS Batch multi-container support
FMI/FMU Support	Yes (auto-generates FMUs)	Yes
MATLAB/Simulink	Yes (tight integration)	Yes (S-function integration)
Real-Time / HIL	Yes	Yes
Industry Use	110+ OEMs and Tier 1s	Widely used in AV/ADAS industry

CHAPTER 3

Calsim Technology Inc.

Note on a Commonly Confused Company Name

Calsim Technology Inc. is a software development company founded in 1997 by Dr. Wei Sheng, headquartered in Calgary, Alberta, Canada. It is important to understand that Calsim Technology is entirely unrelated to automotive simulation. The company provides custom software development and technical outsourcing services to the oil and gas industry and the manufacturing sector.

Note: Do not confuse Calsim Technology (oil & gas software, Calgary) with CalSim 3 (California Department of Water Resources model) or with CarSim (Mechanical Simulation Corporation vehicle dynamics tool). These are three entirely distinct products from unrelated organisations.

PART II

Synthetic Data Generation

Automotive Applications, Tools & Challenges

CHAPTER 4

Introduction to Synthetic Data

Automotive Applications, Toolchains & Challenges

4.1 Why Synthetic Data?

Synthetic data refers to computer-generated data designed to mimic real-world data with high fidelity. In the automotive industry, it has emerged as a critical enabler for developing and validating AI-based systems — particularly ADAS and autonomous driving — where the volume, variety, and quality of training data directly determines system performance and safety.

Collecting and labelling real-world driving data at the scale required to train robust perception systems is extremely expensive. Industry estimates suggest that data collection and annotation can account for up to 70% of total ADAS development costs. Synthetic data generation allows teams to create an almost unlimited amount of precisely labelled data on demand, at a fraction of the cost.

- Eliminates privacy concerns — no real drivers, pedestrians, or licence plates captured
- Enables coverage of rare, dangerous, or edge-case scenarios impractical to capture in the real world
- Provides pixel-perfect ground truth labels automatically — no manual annotation
- Allows early development before physical prototypes exist
- Supports rapid iteration: change a parameter, regenerate thousands of data samples
- Enables systematic design of experiments across scenario variables

4.2 Automotive Applications

Perception System Training

Machine learning models for object detection, lane detection, depth estimation, and semantic segmentation require large, diverse labelled datasets. Synthetic data from tools like CarMaker provides this with automatic ground truth labels at pixel level.

Sensor Development and Calibration

Synthetic data assists in developing and calibrating sensors including lidar, radar, cameras, and ultrasonic sensors. Physics-based sensor models replicate noise, field of view, range falloff, and environmental interference.

Anomaly Detection and Fault Diagnosis

By generating synthetic data representing fault conditions — sensor failures, unusual object behaviours, adverse road conditions — engineers can train anomaly detection algorithms without inducing dangerous physical conditions.

Driver and Occupant Monitoring

Synthetic data generates diverse driver and passenger appearance datasets for in-cabin monitoring systems, covering gaze direction, head pose, emotion, lighting conditions, and camera perspectives.

Powertrain and Thermal Validation

In the eDU context, synthetic drive cycle data can be generated to represent customer usage patterns — sport driving, towing, repeated acceleration events, mountain descent — providing thermal loading profiles for CLS sizing.

4.3 Tools and Platforms

Tool / Platform	Type	Primary Use
CarMaker (IPG Automotive)	Automotive Simulation	Scenario-based synthetic data, ADAS
CARLA	Open-Source Sim	Autonomous driving, ML training
Unreal Engine	Game Engine	High-fidelity photorealistic rendering
Unity	Game Engine	Flexible simulation environment
MATLAB/Simulink	System Simulation	Powertrain, thermal, control systems
SUMO	Traffic Simulation	Traffic flow and interaction modelling
AWS Batch + CarMaker	Cloud HPC	Large-scale parallel data generation
NVIDIA Drive Sim	GPU-based Sim	Sensor simulation, AV validation

4.4 Challenges of Synthetic Data

The Reality Gap (Sim-to-Real Transfer)

The most significant challenge with synthetic data is the reality gap — the difference between how the simulation renders the world and how real sensors perceive it. A model trained exclusively on synthetic data may fail when exposed to real-world inputs due to differences in texture, lighting, object placement, sensor noise, and physical dynamics.

Mitigation Strategies

- Domain randomisation — vary simulation parameters (textures, lighting, noise) to force the model to learn robust features

- Domain adaptation — use techniques such as GANs to translate synthetic images into more realistic-looking outputs
- Hybrid datasets — combine synthetic and real data in training to anchor the model in real-world distributions
- Physics-based sensor models — invest in high-fidelity sensor simulation that accurately replicates noise characteristics
- Continuous validation — regularly benchmark against held-out real-world test sets

Note: At JLR, synthetic data from CarMaker simulations is used in combination with real dyno and vehicle data to validate thermal models of the eDU CLS, especially for drive cycle thermal soak scenarios that would be expensive to replicate physically.

PART III

eDU CLS Stage 1 — System Fundamentals

Cooling & Lubrication Architecture | Oil & Coolant Circuits | Thermal Limits

CHAPTER 5

eDU Cooling Architecture Overview

Two-Circuit Strategy | Heat Paths | System Interactions

5.1 The eDU as a System

An Electric Drive Unit (eDU) integrates three major sub-assemblies into a single compact housing: the AC permanent-magnet synchronous e-motor, the high-voltage power inverter (typically a three-phase IGBT or SiC MOSFET bridge), and a single-speed reduction gearbox with integrated differential. The eDU converts electrical energy from the HV battery into mechanical torque delivered to the driven wheels.

Managing heat within this integrated unit is one of the primary engineering challenges. Each sub-assembly generates heat continuously during operation, and the thermal performance of the CLS directly determines the sustained power output, the frequency of thermal derating events, component longevity, and ultimately the driving experience delivered to the customer.

5.2 Cooling Circuits and Heat Paths

Stator Cooling Jacket

The e-motor stator is the dominant heat source in the eDU. Copper losses in the stator windings (I^2R heating) and iron losses in the lamination stack both generate significant heat. A coolant jacket — a precision-machined or cast channel surrounding the stator outer diameter — carries a water-glycol mixture to extract this heat. The jacket is part of a low-temperature coolant circuit shared with the inverter cold plate and the oil-to-water heat exchanger.

Rotor Oil-Spray System

The rotor and end-windings are cooled by direct oil impingement. Oil is pumped under pressure through internal galleries in the rotor shaft and sprayed onto the end windings via nozzles or holes at the shaft ends. This provides effective direct cooling of the hottest winding locations and lubricates the rotor bearings simultaneously. The oil then drains by gravity back to the sump.

Inverter Cold Plate

The power inverter generates switching losses and conduction losses in the IGBT or SiC MOSFET power modules. A machined cold plate — a flat heat exchanger — is bolted directly to the power module baseplate, conducting heat into the water-glycol circuit. The thermal resistance of this interface is critical: even a few degrees of additional resistance can push IGBT junction temperatures above the derating threshold under peak torque conditions.

Oil-to-Water Heat Exchanger

Oil circulating through the lubrication and rotor-spray circuit absorbs heat from gearbox churning, bearing friction, and rotor spray. This heat must be rejected back into the vehicle thermal management system. The oil-to-water heat exchanger (OWC) provides the thermal interface between the oil circuit and the coolant loop. The effectiveness of the OWC is a key lever in controlling oil sump temperature.

5.3 Heat Source Summary

Heat Source	Cooling Medium	Heat Rejection Path
Stator windings (copper loss I^2R)	Water-glycol	Stator jacket → VTMS radiator
Stator laminations (iron loss)	Water-glycol	Stator jacket → VTMS radiator
Rotor / end-windings	Oil (spray)	Oil → OWC → VTMS
Rotor bearings	Oil (splash/spray)	Oil → OWC → VTMS
Inverter (IGBT/SiC)	Water-glycol	Cold plate → VTMS radiator
Gearbox / differential	Oil (splash)	Oil → OWC → VTMS

5.4 Two Primary Thermal Loops — Fluid Properties

Parameter	Oil Circuit	Coolant Circuit
Fluid	ATF / EV Oil (0W-20)	Water-Glycol 50/50
Density	~870 kg/m ³	~1060 kg/m ³
Specific Heat (Cp)	~2000 J/kg·K	~3600 J/kg·K
Typical Flow	2–10 L/min	10–25 L/min
Temp Range	–40°C to 130°C	–40°C to 115°C
Pump Type	Electric Oil Pump (EOP)	PWM Electric Water Pump
Heat Sink	Oil-to-Water HEX → Coolant	Vehicle Radiator / Chiller
Key Interfaces	Windings, bearings, gears	Stator housing, cold plate

CHAPTER 6

eDU Lubrication Architecture

Oil Sump | Pick-Up Pipe | Oil Pump | Galleries | Splash Lubrication

6.1 Lubrication Circuit Overview

The lubrication system serves two simultaneous functions in the eDU: it provides a hydrodynamic film between all sliding and rolling contact surfaces to prevent wear and seizure, and it acts as a secondary heat transfer medium, absorbing heat from the gearbox and rotor and carrying it to the oil-to-water heat exchanger.

6.2 Main Components

Oil Sump

The sump is the oil reservoir at the base of the eDU housing. It collects all oil draining from the gearbox, differential, rotor spray, and bearings. Sump design must ensure adequate oil volume for cold-start priming, sustained supply under dynamic tilt (off-road and cornering), and sufficient thermal mass to buffer short-duration peak thermal events.

Pick-Up Pipe

The pick-up pipe draws oil from the sump into the inlet of the oil pump. Its inlet is submerged below the design oil level. Correct placement is critical: if the inlet is too close to the surface, dynamic oil movement under lateral acceleration or gradient can expose it to air, causing pump cavitation and momentary oil starvation. A strainer at the inlet prevents coarse debris ingestion.

Oil Pump

The oil pump pressurises the oil to force it through the lubrication and cooling galleries. It may be mechanically driven from the gearbox input shaft or electrically driven (independent of motor speed). Flow rate and pressure are the primary outputs.

Oil Filter

A full-flow filter protects oil galleries, bearings, and nozzles from particulate contamination. A bypass valve opens when the filter is blocked or cold-viscosity oil cannot flow through — preventing catastrophic starvation at the cost of unfiltered oil reaching critical surfaces.

Oil Galleries

Drilled or cast passages distribute pressurised oil from the pump to: main shaft bearings, planet gear bearings in the gearbox, differential bearings, and the rotor shaft for end-winding spray.

Gear and Differential Splash Lubrication

In addition to pressure-fed galleries, gear meshes and the differential are lubricated by oil splash — the rotation of gear teeth dips into the oil level and flings oil onto meshing surfaces and bearings. Splash lubrication dominates at low speeds; pressure lubrication ensures supply at high speeds where dip contact

time is minimal.

CHAPTER 7

Oil Pump — Types, Operating Envelope & Failure Modes

Mechanical vs Electric | Flow vs Speed | Failure Analysis

7.1 Pump Types

Mechanically Driven Oil Pump

A mechanically driven pump is coupled to the eDU input shaft through a fixed gear ratio. Pump speed, flow rate, and output pressure are therefore directly proportional to motor speed. This is simple and reliable — no additional power electronics required — but it cannot provide flow when the vehicle is stationary (e.g. during stationary high-current charging, thermal soak, or creep). During cold start, the pump runs slow until motor speed rises.

Electrically Driven Oil Pump (EOP)

An EOP is driven by its own dedicated DC brushless motor controlled by the VCU. It can operate independently of main motor speed, providing full flow during stationary thermal events, during engine-off coasting, and for pre-lubrication before a cold start drive-off. Modern eDU designs often use an EOP as the primary or sole pump. The trade-off is additional cost, mass, and packaging complexity.

7.2 Flow Rate vs Speed Curves

A positive displacement pump (gear pump or gerotor) produces a flow rate that is nearly linear with pump speed, with a slight rollback at high back-pressure. The circuit resistance (filter, galleries, heat exchanger) creates a system curve — the operating point is the intersection of pump curve and system curve.

- Cold start (–30°C): high viscosity → high system resistance → low flow → risk of starvation in critical galleries
- Warm operation (80–100°C): low viscosity → low resistance → high flow → risk of pump over-speed relief valve opening
- Peak load: maximum heat generation requires maximum flow; pump must be sized for this condition without exceeding mechanical limits

7.3 Oil Pump Failure Modes

Failure Mode	Cause	Effect	Detection
No flow	Pump seizure, coupling failure	Bearing seizure, gear damage	Oil pressure sensor low
Low flow	Wear, cold viscosity, filter blockage	Inadequate cooling/lubrication	Pressure low, temp high
Cavitation	Air ingestion via pick-up pipe	Noise, flow pulsation, wear	Pressure oscillation
Internal leakage	Worn gear tips or rotor	Reduced pressure at load	Pressure low at speed
Relief valve stuck open	Contamination	Pressure capped, flow diverted	Pressure low despite pump running

CHAPTER 8

Coolant Flow Rate

Sizing Logic | Key Equations | Drivers of the Requirement

8.1 How to Size Coolant Flow Rate

The required coolant flow rate through the stator jacket and inverter cold plate is determined by the heat balance equation:

Coolant Heat Balance Equation

$Q = m_{\text{dot}} \times C_p \times \Delta T$

Where:

Q = heat to be rejected (W)
 m_{dot} = mass flow rate (kg/s)
 C_p = specific heat capacity of coolant (~3,500 J/kg.K for 50% WG mix)
 ΔT = allowable coolant temperature rise across component (K)

Coolant sizing (combined EDU loop):
 $Q_{\text{coolant}} = (P_{\text{inverter_loss}} + P_{\text{stator_jacket_loss}}) / (\rho_{\text{wg}} \times C_{p_wg} \times \Delta T)$
 $\rho_{\text{wg}} = 1060 \text{ kg/m}^3$ $C_{p_wg} = 3600 \text{ J/kg.K}$ Typical ΔT : 5–15°C

8.2 Key Drivers of the Requirement

- Peak motor copper + iron losses can exceed 10–15 kW in a high-performance eDU
- Inverter switching losses may add a further 3–5 kW
- Allowable ΔT across stator jacket is typically 5–10°C to maintain winding temperature margin below the insulation class limit
- This dictates flow rates in the range of 5–15 litres/minute depending on eDU power rating
- The VTMS coolant pump must deliver this flow while overcoming the pressure drop through the jacket, cold plate, pipework, and heat exchanger

Typical Flow Targets by Component

Component	Typical Flow (L/min)	Max Inlet Temp	Typical delta-T
Inverter cold plate	8–15	65°C	5–10°C
Stator housing jacket	10–20	70°C	8–15°C
Combined EDU coolant loop	15–25	70°C	10–15°C

CHAPTER 9

Oil Flow Rate

Lubrication Adequacy vs Cooling Adequacy | Competing Demands

9.1 Lubrication Adequacy vs Cooling Adequacy

Oil flow rate must satisfy two independent requirements simultaneously: lubrication adequacy and cooling adequacy. These can pull in opposite directions.

Lubrication Adequacy

Every bearing and gear mesh has a minimum required oil flow to maintain a hydrodynamic film and flush away heat and debris. This is a minimum threshold — below it, wear and eventually seizure occur. The requirement is relatively independent of temperature once the oil is warm.

Cooling Adequacy

The rotor oil-spray system must deliver sufficient oil mass flow to limit end-winding temperatures under peak electrical loading. This requirement scales with heat generation — at high torque, more oil flow is needed to keep winding temperatures below the insulation class limit.

9.2 Competing Demands

- At low motor speeds, lubrication requirements dominate — the EOP must maintain minimum gallery pressures even when the main shaft pump produces little flow
- At high motor speeds and loads, cooling requirements dominate — flow must be sufficient to remove winding heat fast enough
- Cold start creates a transient competing demand: high viscosity limits flow while the motor is cold, yet lubrication adequacy must still be met
- Oil temperature management at the OWC must balance: higher oil temperature reduces viscosity (good for flow, bad for film strength)

9.3 Oil Pump Sizing Logic

Oil Flow Requirement Calculation

$$Q_{\text{required}} = P_{\text{loss}} / (\rho \times C_p \times \Delta T_{\text{allowable}})$$

Q = volumetric flow rate (m^3/s or L/min)

P = heat generated in component (W)

ρ = oil density ($\sim 870 \text{ kg}/\text{m}^3$ for ATF)

C_p = specific heat capacity ($\sim 2000 \text{ J}/\text{kg}\cdot\text{K}$)

ΔT = allowable temperature rise across component (K)

Flow demand is driven by heat rejection requirement – NOT by motor speed alone.
The EOP is fully decoupled, allowing independent thermal and lubrication control.

CHAPTER 10

Oil Sump Design

Oil Volume | Geometry | Baffling | Temperature Management

10.1 Design Considerations

- Oil volume: must be large enough for cold-start priming lag, steady-state operation, and oil thermal buffer capacity
- Sump geometry: must not allow oil level to drop below the pick-up pipe inlet under all tilt angles (off-road gradients, cornering, braking, acceleration)
- Baffling: internal baffles prevent oil surge exposing the pick-up pipe inlet
- Drain-back channels: oil from rotor spray and bearing drain-off must return efficiently to sump without excessive residence time in hot zones
- Breather: sump must be vented to prevent pressure build-up while preventing contamination ingress
- Magnetic drain plug: captures ferrous wear particles during service life, used as a condition monitoring tool during PDI and service

10.2 Sump Temperature

Oil sump temperature is the thermal baseline of the lubrication system. It must remain below the oil degradation limit (typically 120–130°C for mineral or synthetic gear oil) under the worst-case sustained drive cycle. At high sump temperatures, oil viscosity falls below the minimum required for hydrodynamic lubrication, accelerating wear. Sump temperature is managed by controlling oil flow rate through the OWC and the coolant temperature in the vehicle circuit.

CHAPTER 11

Pick-Up Pipe

Function | Air Ingestion Risk | Cold Start Starvation

11.1 Function and Placement

The pick-up pipe is the inlet to the oil circuit. It connects the sump to the inlet of the oil pump. Its inlet must remain permanently submerged below the oil surface under all operating conditions to prevent air ingestion.

11.2 Air Ingestion Risk

Air ingestion at the pick-up pipe is a serious failure mode. Even momentary air ingestion causes pump cavitation — the pump loses its priming, flow rate drops precipitously, and oil pressure collapses. Bearings and

gear meshes that rely on pressure-fed oil can seize within seconds. The risk is highest during:

- High lateral acceleration (cornering) — oil shifts to one side of the sump
- Steep gradients (off-road) — oil pools at the low end of the sump
- Combined gradient and cornering — worst-case multi-axis tilt
- Low oil level due to oil consumption or incorrect fill at service
- Dynamic deceleration — oil surges forward under heavy braking

11.3 Cold Start Starvation Risk

At very low temperatures, oil viscosity is extremely high. The oil pump must generate sufficient suction to draw viscous oil through the pick-up pipe and strainer. If the strainer is partially blocked by wax precipitation or debris, the pressure drop across it may exceed pump suction capacity, causing starvation even if oil level is adequate.

CHAPTER 12

Winding Temperatures

Insulation Classes | Thermal Limits | Measurement Methods

12.1 Why Winding Temperature Matters

The stator winding insulation system is the life-limiting component of the e-motor. Insulation is a polymer film applied to the copper conductor. At elevated temperatures, the film undergoes accelerated thermal ageing, reducing dielectric strength. Beyond the thermal class limit, insulation degrades rapidly, leading to inter-turn short circuits, phase-to-phase shorts, or winding-to-ground faults — all of which are catastrophic motor failures.

12.2 Insulation Classes

Insulation Class	Maximum Winding Temperature	Typical Application
Class F	155°C	Standard industrial motors
Class H	180°C	High-performance automotive e-motors
Class N / partial discharge resistant	200°C+	Premium EV motors (800V systems)

Most automotive eDU motors use Class H insulation. The allowable continuous temperature is 180°C, but motors are typically derated to 160°C continuous to maintain a margin against hotspot effects and measurement uncertainty.

12.3 Measurement and Management

- NTC (Negative Temperature Coefficient) thermistors embedded between stator windings provide direct temperature measurement with ~1–2°C accuracy

- Thermistor locations are chosen to represent the hottest winding location or the average winding temperature, depending on the control strategy
- The VCU reads winding temperature and applies current limiting (derating) when the threshold is approached
- Thermal models (lumped-parameter or FEA-based) are used to estimate winding temperatures at un-sensed locations between the thermistors

CHAPTER 13

Bearing Temperatures

Failure Modes | Lubrication Starvation | Monitoring Strategies

13.1 Failure Modes from Overtemperature

- Oil film breakdown — at high temperatures, oil viscosity drops below the minimum for hydrodynamic film formation; metal-to-metal contact increases wear rate dramatically
- Grease decomposition — in sealed bearings with grease packing, high temperature causes grease to oxidise, lose viscosity, and carbonise
- Differential thermal expansion — bearing clearances can close up at high temperature, increasing contact stress and frictional heating in a positive feedback loop leading to seizure
- Cage fracture — at high speeds and temperatures, bearing cage fatigue accelerates, potentially causing cage fracture and catastrophic bearing failure
- Raceway spalling — fatigue of the bearing raceway under combined thermal and mechanical load cycling

13.2 Lubrication Starvation Effects

Bearing temperature and lubrication starvation are closely linked. If oil supply to a bearing is interrupted — due to pump failure, blocked gallery, or excessive air ingestion — bearing temperature rises rapidly. The sequence is: starvation → temperature rise → film breakdown → surface damage → seizure. The time to seizure depends on bearing load, speed, and residual oil film in the contact zone.

13.3 Monitoring Strategies

- Direct NTC or PT100 temperature sensors on bearing housings
- Indirect monitoring via oil sump temperature and oil pressure sensors
- Acoustic emission monitoring — bearing damage generates characteristic high-frequency acoustic signatures detectable with accelerometers
- Oil particle count — metal debris in oil is an early indicator of bearing surface damage, detectable by inline particle sensors or oil sampling

CHAPTER 14

Churning Losses

Definition | Key Influences | Link to Efficiency & Thermal Load

14.1 What are Churning Losses?

Churning losses arise when rotating gears, shafts, and bearings agitate the oil in the sump. The viscous drag of oil on rotating components dissipates mechanical energy as heat. Unlike gear mesh losses (which occur only at meshing contacts), churning losses scale with the volume of oil displaced per revolution and the oil viscosity.

14.2 Key Influences

- Oil viscosity — higher viscosity (low temperature, wrong oil grade) → higher churning losses; this is why gear oils are formulated with viscosity modifiers
- Oil level — higher oil level in the sump → more oil contact with rotating components → higher churning losses; some eDUs use oil level control valves to lower the level at high speed
- Gear ratio — higher gear ratio means the differential and output gears rotate faster relative to road speed; at high vehicle speeds, churning power loss can exceed 1–2 kW in a poorly optimised design
- Shrouding — shrouds or baffles around gears physically separate them from the bulk oil, reducing drag
- Operating temperature — as oil warms up, viscosity falls and churning losses reduce; this is why eDU efficiency improves after warm-up

14.3 Link to Efficiency and Thermal Load

All churning losses appear as heat in the oil circuit. This directly loads the oil-to-water heat exchanger and raises sump temperature. In a worst-case cold start high-speed scenario, churning losses can drive sump temperature up rapidly. Understanding and minimising churning losses is therefore both an efficiency target (range) and a thermal management requirement (CLS sizing).

PART IV

eDU CLS Stage 2 — Thermal & Efficiency Engine

Losses | Drive Cycles | Derating Strategy | Gear Ratios

CHAPTER 15

Losses and Efficiency in the eDU

Copper, Iron, Switching, Mechanical & Churning Losses

15.1 Loss Sources

Loss Type	Location	Scales With	Heat Path
Copper losses (I^2R)	Stator windings	Current squared	Stator jacket → coolant
Iron losses (eddy + hysteresis)	Stator/rotor laminations	Frequency × flux density	Stator jacket → coolant
Switching losses	IGBT/SiC switches	Switching frequency	Inverter cold plate → coolant
Conduction losses	IGBT/SiC on-state resistance	Current squared	Inverter cold plate → coolant
Mechanical losses	Bearings, seals, windage	Speed	Oil → OWC
Churning losses	Gears in oil sump	Speed, viscosity	Oil → OWC
Gear mesh losses	Gear contact zones	Load × speed	Oil → OWC

A typical high-performance eDU has a peak efficiency of 93–96% at optimal speed/torque conditions. At partial loads and very low speeds, efficiency can fall to 80–85%. Understanding where losses occur and how they manifest as heat in specific sub-circuits is essential for CLS design.

CHAPTER 16

Drive Cycles

WLTP | NEDC | Customer Duty Cycles | Thermal Loading

16.1 Standard Drive Cycles

Cycle	Region	Max Speed	Character	Thermal Significance
WLTP Class 3	Global/EU	131 km/h	Mixed urban + extra-urban + highway, 23 min	Gold standard — homologation requirement
US06	USA	129 km/h	Aggressive, rapid acceleration	Worst-case peak thermal stress
NEDC	Legacy EU	120 km/h	Gentle, low average speed	Too mild — being phased out
RTS-95	EU	120 km/h	Highway-focused, sustained high speed	Sustained thermal load on rotor
FTP-75	USA	91 km/h	Urban, stop-start	Low-moderate thermal load
Custom OEM	OEM-specific	Variable	Grade climbs, towing, trailer	Worst-case use-case validation

16.2 How Drive Cycles Define Thermal Loading

Each point on a drive cycle maps to a motor operating condition (torque, speed). The motor efficiency map (a 2D torque-speed surface) converts each operating point into a heat generation rate. Integrating heat generation over the cycle gives the thermal energy that must be rejected by the CLS. Peaks in heat generation rate set the instantaneous cooling requirement; integrated heat over sustained high-load segments sets the sustained requirement.

CHAPTER 17

E-Motor Thermal Derating

Trigger Conditions | VCU Response | Vehicle Performance Impact

17.1 What Triggers Derating?

E-motor thermal derating occurs when winding temperatures approach the insulation class limit. The VCU reads the NTC thermistor in the stator and applies a current limit reduction — reducing peak torque — to bring heat generation back within the CLS rejection capacity. Derating is progressive: a graduated reduction in allowable current as temperature rises, rather than a binary cut-off.

17.2 Impact on Vehicle Performance

- Reduced peak torque available at the driven wheels — acceleration performance is compromised during sustained high-performance driving
- In performance vehicles, sustained track use can trigger repeated derating events if the CLS is not sized for the thermal duty
- Customer perception of 'power fade' is often a thermal derating symptom
- Regenerative braking capacity may also be limited by winding temperature during long downhill sections

CHAPTER 18

Inverter Thermal Derating

IGBT Junction Temperature | Derating Curves | Coolant Temperature

18.1 IGBT Junction Temperature Limits

The IGBT (or SiC MOSFET) power modules in the inverter have a maximum junction temperature typically of 150–175°C. The junction temperature is significantly higher than the cold plate temperature due to the thermal resistance chain: junction → baseplate → TIM (thermal interface material) → cold plate → coolant.

18.2 Derating Curves and Coolant Temperature

Inverter suppliers provide derating curves showing allowable continuous current vs. coolant inlet temperature. As coolant temperature rises, available current is reduced to keep junction temperature within limits. At a coolant temperature of 65°C, the available current may be only 70–80% of the value at 25°C coolant. This makes the VTMS coolant temperature a direct control variable for inverter performance headroom.

CHAPTER 19

Thermal Derating Strategy

CLS Performance Sets the Boundary | Margin Management

19.1 CLS Performance Sets the Derating Boundary

The derating thresholds for motor and inverter are set in software, but the actual derating frequency and depth experienced in a real vehicle is determined by CLS hardware performance. A CLS that maintains lower coolant and oil temperatures under the same drive cycle will allow the motor and inverter to operate at higher performance levels for longer before derating activates.

19.2 Margin Management

Systems engineers must define thermal budgets and margins. The thermal budget is the allowable temperature at the winding or junction under the worst-case customer duty cycle. The margin is the difference between the budget and the derating threshold. Margin must account for: measurement uncertainty of the temperature sensor, variation in CLS performance across the production build population, and degradation of cooling performance over vehicle life (e.g. coolant contamination, partial filter blockage).

CHAPTER 20

Gear Ratios and Thermal Load

Motor Operating Point | Losses | Churning at Speed

20.1 Gear Ratio and Motor Operating Point

A single-speed eDU uses a fixed gear ratio between the motor shaft and the driven wheels. The gear ratio selection directly determines the motor speed at any given road speed, and therefore the motor operating point on the torque-speed efficiency map.

- High gear ratio → motor runs at high speed for a given road speed → risk of high iron losses and mechanical losses at motorway cruising
- Low gear ratio → motor runs at low speed → poor efficiency at low torque light-load conditions; large motor size needed for acceptable acceleration
- Optimum gear ratio balances peak torque requirement, maximum speed requirement, and best average efficiency across the customer drive cycle
- Higher gear ratios increase differential and output gear rotational speed, increasing churning losses and bearing temperatures at high road speeds

PART V

eDU CLS Stage 3 — Requirements Engineering

Stakeholder Needs | Cascade | EARS | Interface Requirements

CHAPTER 21

Stakeholder Needs vs System Requirements

Distinction, Transformation & Worked Examples

21.1 The Distinction

A stakeholder need is an expression of what someone wants the system to do, expressed from their perspective, often without specifying how. A system requirement is a precise, verifiable statement of what the system shall do or what constraint it shall meet. The transformation from needs to requirements is a structured engineering activity that must capture intent while making the requirement testable.

Stakeholder Need	Derived System Requirement (example)
Motor must not derate during a full lap of a standard track	The CLS shall maintain winding temperature below 155°C during the standard test track lap
eDU must survive 15-year vehicle life	The CLS shall maintain oil sump temperature below 120°C during the WLTP cycle to prevent
Pass regulatory type approval	The CLS shall support continuous WLTP cycle operation at 40°C ambient without exceeding
Oil change interval manageable at standard service	The CLS oil circuit shall maintain oil quality within Grade 3 degradation index throughout a 2

CHAPTER 22

Requirements Cascade and Traceability

Vehicle → EDU → Motor → CLS → Component

22.1 The Cascade Process

Requirements traceability is the ability to follow a requirement from its origin (stakeholder need or higher-level requirement) through to the system requirements, subsystem requirements, component requirements, and ultimately the verification evidence. Every requirement must have a parent and the derivation logic must be documented.

Example Cascade — Winding Temperature

Requirements Cascade — Winding Temperature Example

Vehicle-level need:
'No performance derating on the consumer drive cycle'

↓

System requirement:
'eDU CLS shall reject motor heat to maintain winding temperature $\leq 155^{\circ}\text{C}$ under customer duty cycle at 40°C ambient'

↓

Subsystem requirement:
'Stator cooling jacket shall achieve a thermal resistance $\leq 0.05\text{ K/W}$ between winding and coolant at nominal flow rate'

↓

Component requirement:
'Jacket flow passage shall produce a delta-T of $\leq 8^{\circ}\text{C}$ across the jacket at 8 litres/minute coolant flow'

22.2 Vehicle → EDU Level Cascade Example

0–100 km/h Acceleration Requirement Cascade

VEHICLE REQUIREMENT: 0–100 km/h in 5.5 seconds

Step 1 – Average acceleration: $a_{\text{avg}} = (100/3.6) / 5.5 = 5.05\text{ m/s}^2$

Step 2 – Peak tractive force ($M=2200\text{ kg}$): $F = 2200 \times 5.05 + F_{\text{aero}} + F_{\text{rolling}} \approx 12,000\text{ N}$

Step 3 – Wheel torque ($R_{\text{wheel}} = 0.35\text{ m}$): $T_{\text{wheel}} = 12,000 \times 0.35 = 4,200\text{ Nm}$

Step 4 – EDU output torque (final drive 9.5:1, $\eta=0.97$):
 $T_{\text{EDU}} = 4200 / (9.5 \times 0.97) = 456\text{ Nm} \leftarrow \text{EDU REQUIREMENT}$

Step 5 – EDU output speed at 100 km/h:
 $\omega_{\text{wheel}} = (100/3.6) / 0.35 = 79.4\text{ rad/s}$
 $\omega_{\text{EDU}} = 79.4 \times 9.5 = 754\text{ rad/s} = 7,200\text{ RPM} \leftarrow \text{EDU REQUIREMENT}$

CHAPTER 23

Functional vs Non-Functional Requirements

Definitions & Worked Examples for the CLS

23.1 Functional Requirements

Functional requirements define what the system shall do:

- The CLS shall circulate oil to all eDU bearing locations within 5 seconds of eDU start-up at -30°C ambient temperature
- The CLS shall reject a minimum of 12 kW of heat from the stator during continuous rated motor operation
- The CLS shall provide oil flow to the rotor end-winding spray nozzles at all motor operating speeds above 500 rpm
- The coolant pump shall maintain a minimum flow rate of 6 l/min through the stator cooling jacket at all operating conditions within the system boundary
- The oil-to-water heat exchanger shall limit oil inlet temperature to the pump to below 110°C during sustained customer duty cycle operation
- The CLS shall provide a warning to the VCU if oil pressure falls below the minimum defined threshold for more than 2 seconds

23.2 Non-Functional Requirements

Non-functional requirements define quality attributes of how the system shall perform:

- The CLS shall have a design life of 300,000 km or 15 years, whichever is sooner
- The CLS oil pump shall produce no more than 65 dB(A) noise at 1 metre distance at any operating speed within the defined speed range
- The CLS shall be serviceable (oil drain and fill) using standard workshop tools in a maximum of 30 minutes
- The CLS shall meet IP6K9K ingress protection rating for all external electrical connectors (temperature sensors, pressure sensors, EOP motor connector)
- The CLS oil circuit shall not exhibit external oil leakage exceeding 0 ml/1000 km under any combination of temperature, pressure, and vibration within the design envelope
- The total parasitic power consumed by the CLS (EOP motor + coolant pump) shall not exceed 400 W at the WLTP average operating condition

CHAPTER 24

EARS Syntax

Easy Approach to Requirements Syntax — Five Patterns

24.1 What is EARS?

EARS (Easy Approach to Requirements Syntax) is a structured natural-language template for writing requirements that minimises ambiguity. It uses keywords to capture the trigger, precondition, and required response of the system. The five main patterns are: Ubiquitous, Event-driven, State-driven, Option-driven, and Unwanted Behaviour.

EARS Pattern	Template	CLS Example
Ubiquitous	The [system] shall [action]	The CLS shall provide oil to all gearbox bearings
Event-driven	WHEN [trigger] the [system] shall [action]	WHEN oil pressure drops below 1.5 bar for > 2s, the CLS shall signal a DTC to the VCU
State-driven	WHILE [state] the [system] shall [action]	WHILE the eDU is in stationary charging mode, the EOP shall maintain a minimum oil level
Option-driven	WHERE [feature] the [system] shall [action]	WHERE high-performance mode is enabled, the CLS shall prioritise maximum cooling capacity
Unwanted behaviour	IF [condition] THEN the [system] shall [action]	IF winding temperature exceeds 160°C THEN the VCU shall reduce motor current

CHAPTER 25

Interface Requirements

CLS Interfaces with Gearbox | E-Motor | Inverter | VCU | VTMS

25.1 Interface Definition

CLS interface requirements define what the CLS shall provide to, and what it shall receive from, each connected system. These are critical for system integration and prevent interface mismatches discovered late in the programme.

Interface	CLS provides to...	CLS receives from...
Gearbox	Pressurised oil to bearing galleries and gear splash zone	Oil temperature, oil drain back to sump, debris load on filter
E-Motor	Oil flow to rotor spray nozzles and stator jacket cooling	Winding temperature signal, heat load (copper + iron losses)
Inverter	Coolant flow through cold plate at defined inlet temperature	GBT junction temperature signal, switching loss heat
Vehicle Coolant Circuit (VTMS)	Coolant return at elevated temperature	Coolant supply at defined inlet temperature and flow rate
VCU	Oil pressure signal, oil temperature, winding temp, cooling fan speed	500 bar/pressure command, coolant pump speed command, derating
HV Battery Cooling Loop	Thermal isolation (separate circuit)	Shared VTMS chiller capacity at defined conditions

CHAPTER 26

Pressure and Temperature Sensor Requirements

Location | Accuracy | Response Time | Functional Safety

Sensor	Location	Accuracy	Response Time	Functional Safety Note
Oil pressure sensor	Downstream of oil pump outlet	± 0.1 bar	< 100 ms	QM — provides early warning; not directly safety-critical
Oil sump temperature	Oil sump, submerged	$\pm 2^{\circ}\text{C}$	< 500 ms	QM — used for oil condition monitoring
Winding temperature (NTC)	Between stator windings	$\pm 2^{\circ}\text{C}$	< 200 ms	ASIL B — used for derating control; loss leads to insulation degradation
Coolant inlet temperature	Before stator jacket inlet	$\pm 1^{\circ}\text{C}$	< 500 ms	QM — used for thermal model and flow control
Coolant outlet temperature	After stator jacket outlet	$\pm 1^{\circ}\text{C}$	< 500 ms	QM — used with inlet to calculate heat rejection
Inverter cold plate temp	Cold plate outlet or IGBT baseplate	$\pm 2^{\circ}\text{C}$	< 100 ms	ASIL A — used for inverter derating control

PART VI

eDU CLS Stage 4 — System Architecture & Design

Functional Architecture | Physical Architecture | Context | MBSE

CHAPTER 27

Functional Architecture

Heat Rejection | Flow Control | Lubrication Distribution | Thermal Monitoring

The functional architecture of the CLS describes what the system must do, expressed as a hierarchy of functions, without yet specifying which physical component performs each function.

F1 — Heat Rejection

Transfer thermal energy from all eDU sub-assemblies to the vehicle thermal management system

- F1.1 — Reject stator heat via coolant circuit
- F1.2 — Reject inverter heat via coolant circuit
- F1.3 — Reject gearbox and rotor heat via oil circuit and OWC

F2 — Flow Control

Regulate coolant and oil flow rates to match thermal demand

- F2.1 — Control EOP speed as a function of VCU command
- F2.2 — Control coolant pump speed as a function of thermal demand
- F2.3 — Regulate oil pressure via relief valve

F3 — Lubrication Distribution

Deliver oil at sufficient pressure and flow to all bearing and gear lubrication points at all times

- F3.1 — Pressurise and filter oil supply
- F3.2 — Distribute oil to bearings via galleries
- F3.3 — Distribute oil to rotor spray nozzles

F4 — Thermal Monitoring

Measure and report temperatures and pressures to VCU

- F4.1 — Monitor winding temperature

- F4.2 — Monitor oil pressure and temperature
- F4.3 — Monitor coolant inlet and outlet temperatures

CHAPTER 28

Physical Architecture

Allocating Functions to Physical Components

The physical architecture allocates functions to specific hardware components:

Function	Physical Component
Reject stator heat via coolant	Stator cooling jacket (cast into motor housing)
Reject inverter heat via coolant	Inverter cold plate (bolted to IGBT baseplate)
Reject gearbox/rotor heat via oil	Oil-to-water heat exchanger (bolted to eDU housing)
Control EOP speed	Electric oil pump + VCU EOP driver circuit
Control coolant pump speed	VTMS variable-speed coolant pump
Regulate oil pressure	Pressure relief valve (integral to pump or housing)
Pressurise and filter oil	Oil pump + oil filter assembly
Distribute oil to bearings	Drilled oil galleries in shafts and housing
Distribute oil to rotor spray	Rotor shaft internal passages + spray nozzles
Monitor winding temperature	NTC thermistors embedded in stator (×2 minimum)
Monitor oil pressure	Pressure sensor on pump outlet gallery
Monitor oil temperature	NTC sensor in oil sump
Monitor coolant temperatures	NTC sensors at jacket inlet and outlet

CHAPTER 29

Context Diagram

System Boundary | All External Interfaces

A context diagram establishes the system boundary of the CLS and identifies all external systems with which it interfaces. The CLS sits within the eDU mechanical and electrical boundary.

- Vehicle Thermal Management System (VTMS) — provides and receives coolant flow

- Vehicle Control Unit (VCU) — sends pump commands, receives temperature and pressure data, issues derating thresholds
- E-Motor sub-system — CLS cools; the motor is the primary heat source
- Power Inverter sub-system — CLS cools; receives IGBT temperature feedback
- Gearbox/Differential sub-system — CLS lubricates and cools; gearbox generates churning and mesh losses
- HV Battery system — shares VTMS capacity (indirect interface via VTMS)
- Service technician (human actor) — oil drain, fill, filter change during service
- Ambient environment — drives worst-case thermal loading (40°C ambient)

CHAPTER 30

Operating Modes

Cold Start | Normal Drive | Sport | Regen | Charging | Thermal Soak

Mode	CLS Behaviour
Cold Start (< -10°C)	EOP runs at minimum speed; oil viscosity limits flow. Winding heat-up assisted. Bypass thermostat keeps coolant in
Normal Drive (warm)	EOP at steady-state speed. Coolant pump at moderate speed. Temperatures within normal operating band. Thermo
Sport / High Torque	EOP at maximum speed. Coolant pump at maximum speed. Winding and IGBT temps monitored closely. Derating th
Regen Braking	Motor acts as generator. Copper losses still significant. Winding temperature continues to rise if sustained. VSC min
Stationary Charging	Motor shaft stationary — no mechanical oil pump contribution. EOP operates independently to circulate oil for bearin
Thermal Soak (post-drive)	eDU unpowered but hot. EOP may be commanded to run briefly for post-drive bearing protection. No active torque -

CHAPTER 31

MBSE Basics — Block Definition Diagrams

SysML BDD | CLS as the Example Model

31.1 What is a BDD?

A Block Definition Diagram (BDD) in SysML is used to define the types of blocks (components or systems) in a system model and their structural relationships. It answers the question 'what kinds of things exist in this system?' — not where they are or how they connect.

31.2 CLS Block Structure

- CLS as the top-level block with its properties (oil volume, coolant flow capacity)

- Oil Circuit as a sub-block with its own properties (flow rate, pressure, temperature range)
- Coolant Circuit as a sub-block (flow rate, delta-T, pump power)
- OWC as a shared interface block between oil and coolant circuits
- Sensor cluster as a block containing pressure sensor, temperature sensors
- EOP as a sub-block with properties: rated flow, rated pressure, motor power
- Oil Filter as a sub-block with properties: micron rating, bypass pressure
- Associations showing composition relationships (CLS is composed of oil circuit + coolant circuit + sensor cluster)

Note: In a real MBSE model, each block has defined value properties (with units and types), flow ports (oil in/out, coolant in/out, electrical power), and interface blocks connecting the ports. Tools such as Cameo, Rhapsody, or Capella are used to create and maintain these models.

PART VII

eDU CLS Stage 5 — Verification & Validation

Methods | Levels | Rigs | Dyno | Acceptance Criteria | Traceability

CHAPTER 32

The Four V&V Methods

Test | Analysis | Inspection | Demonstration

Method	Definition	CLS Application	When Best Used
Test	Execute the system or component and measure output against pass/fail criteria; measure physical hardware exists	Measure output against pass/fail criteria; measure physical hardware exists	When measurement is required to verify design
Analysis	Use models, calculations, or simulation to demonstrate compliance	Calculate heat rejection capacity from flow rate and inlet temperature	When detailed simulation is required to verify design
Inspection	Visual or dimensional check against drawings or specifications	Inspect specified inlet clearance to sump	Physical properties, dimensions, materials, and tolerances
Demonstration	Operate the system and observe behavior	Start the Detailed measurement	Qualitative behavior during functional response;

CHAPTER 33

V&V Levels

Component | Integration | System | Vehicle

Unit / Component Level

Individual components (oil pump, heat exchanger, filter, sensor) tested or inspected against component specifications on dedicated component test rigs.

Integration Level

Oil circuit or coolant circuit tested as an assembled sub-system. Interfaces between components verified. No motor or inverter heat load applied — passive circuit testing.

System Level

Complete eDU CLS tested on an eDU dyno rig with real motor, inverter, and gearbox. Actual heat generation from motor losses applied. All sensors and control loops active.

Vehicle Level

eDU CLS operated in the full vehicle on a chassis dyno or proving ground. VTMS interaction verified. Ambient conditions controlled. Customer duty cycles executed.

CHAPTER 34

Testing Rigs

Oil Circuit Rig | Thermal Soak | Cold Start | Back-to-Back Dyno | Chassis Dyno

Rig	Description	What is Verified
Oil Circuit Rig	Bench rig replicating the oil hydraulic circuit with a simplified eDU speed, pressure, and temperature	Flow rate, oil speed, pressure, and temperature behaviour, cold start
Thermal Soak Rig	Rig that heats the eDU to operating temperature and provides power, soak time, and power decay	Power delivery, soak time, and power decay strategy effectiveness
Cold Start Rig	Environmental chamber with complete eDU, cooled oil, and cold start pump data; bearing lubrication	Oil-soak, cold start, bearing lubrication, pump priming
Back-to-Back Dyno	Two eDUs connected to a shared dyno frame — one efficiency test, the other controlled cooling	Efficiency test, thermal control, cooling sustained
Chassis Dyno	Full vehicle on a roller dyno in an environmental chamber (40°C to 60°C ambient)	Whole system performance, customer duty cycle derating verification

CHAPTER 35

Dyno Testing for CLS Validation

Controlled Torque-Speed | Key Measurement Channels

35.1 How a Dyno is Used

A dyno test rig applies a controlled torque-speed demand to the eDU and measures its response. For CLS validation, the torque-speed profile follows a representative customer duty cycle or a specific worst-case thermal test sequence. The dyno absorbs the mechanical output power and allows precise control of motor operating point throughout the test.

35.2 Key Measurement Channels

- Winding temperature — NTC thermistor signals (×2 minimum); verify against derating limit
- Bearing temperature — housing temperature sensors at each bearing location
- Oil sump temperature — NTC sensor submerged in sump; verify against oil degradation limit
- Oil pressure — pump outlet pressure; verify minimum pressure at all speeds

- Oil flow rate — flow meter on pump outlet line or heat exchanger bypass
- Coolant inlet temperature — entering the stator jacket and cold plate
- Coolant outlet temperature — exiting the stator jacket and cold plate
- Coolant delta-T — difference between inlet and outlet; used to calculate heat rejection
- Coolant flow rate — measured with an inline flow meter
- Inverter cold plate temperature — IGBT baseplate or module case temperature
- Motor efficiency — calculated from shaft torque, speed, and electrical power input
- EOP current and voltage — to monitor EOP operating point and parasitic power

CHAPTER 36

Writing Acceptance Criteria

Winding Temp | Sump Temp | Cold Plate dT | Bearing Flow | Cold Start

Parameter	Acceptance Criterion	Test Condition
Max winding temperature	≤ 155°C	Customer worst-case duty cycle, 40°C ambient, 3 consecutive repeats
Max oil sump temperature	≤ 120°C	Customer duty cycle, sustained high-torque phase, 40°C ambient
Inverter cold plate delta-T	≤ 12°C between inlet and outlet	Peak inverter power (from efficiency map), nominal coolant flow rate
Min bearing oil flow	≥ 0.3 l/min at each main shaft bearing	Minimum motor speed (500 rpm), minimum oil temperature (80°C)
Max winding temp (post thermal soak)	≤ 170°C	30-min thermal soak at max torque immediately followed by 5-min rest
Oil pressure (cold start)	≥ 1.0 bar at pump outlet	Within 5 seconds of eDU start at -30°C soak

CHAPTER 37

Worst-Case Test Scenarios

Thermal Soak | Cold Start | Repeated Cycles | Regen | Stationary Charging

Thermal Soak at Maximum Torque

Hold the eDU at the maximum continuous torque point for 30–60 minutes with worst-case coolant and oil inlet conditions, to verify the CLS can reject heat at steady state.

Cold Soak + Immediate High-Torque Demand

Soak eDU to -30°C overnight, then immediately apply maximum torque demand. Simulates a cold-morning hard acceleration event. Verifies cold-start lubrication adequacy and bearing survival.

Repeated Drive Cycle Derating Check

Run the customer duty cycle 3–5 consecutive times without cooling recovery between cycles. Simulates a worst-case user repeatedly driving hard. Verifies that cumulative thermal build-up does not cause sustained derating.

Mountain Descent Regen

Sustained regenerative braking on a simulated long downhill, with full regen torque applied for 10+ minutes. Verifies inverter cold plate thermal management under sustained reverse power flow.

Stationary Charging at Ambient Temperature Extreme

eDU stationary, HV charging at maximum rate, ambient 50°C. Verifies EOP and coolant pump adequacy for inverter cooling without shaft rotation.

CHAPTER 38

V&V Traceability Matrix

Linking Requirement → Method → Level → Rig → Pass/Fail Criterion

Req. ID	Requirement Summary	V&V Method	Test Level	Rig / Vehicle	Pass/Fail Criterion
CLS-REQ-001	Winding temp ≤ 155°C on customer test cycle	Test	System	Back-to-back dyno	< 155°C on all thermistors
CLS-REQ-002	Oil pressure ≥ 1.0 bar within 5s at 30°C start	Test	System	Cold start rig	Pressure ≥ 1.0 bar at t ≤ 5s
CLS-REQ-003	Oil sump temp ≤ 120°C on customer test cycle	Test	System	Back-to-back dyno	Sump NTC < 120°C throughout
CLS-REQ-004	Inverter cold plate delta-T ≤ 12°C at peak power	Test + Analysis	System + Unit	Dyno + CFD model	Measured ≤ 12°C; model within 10%
CLS-REQ-005	Min bearing oil flow ≥ 0.3 l/min	Test	Integration	Oil circuit rig	Flow meter reading ≥ 0.3 l/min at 5000rpm
CLS-REQ-006	CLS IP6K9K on electrical connectors	Inspection + Test	Component	IP test bath	No water ingress after 30s high-pressure

PART VIII

eDU CLS Stage 6 — Functional Safety & DFMEA

ISO 26262 | HARA | ASIL | TSRs | Requirements Cascade from Safety

CHAPTER 39

DFMEA vs Process FMEA

Design Failure Mode & Effects Analysis | Scope & Distinction

A Design FMEA (DFMEA) analyses the design of a product or system to identify potential failure modes at component or sub-system level, assess their effects on the system and the customer, evaluate their causes, and drive design improvements or detection controls to mitigate risk.

A Process FMEA (PFMEA) analyses the manufacturing or assembly process to identify where process steps could introduce failures into the product — e.g. wrong torque on an oil fitting, missing O-ring on a sensor port, or contaminated oil fill.

Attribute	DFMEA	PFMEA
Focus	Product design	Manufacturing / assembly process
Team	Design and systems engineers	Manufacturing and quality engineers
Failure modes	Component fails to perform its design function	Process step produces a defect in the product
Controls	Design changes, tolerances, redundancy, detection sensors	Process controls, inspection, mistake-proofing (poka-yoke)
When	During design phase, before design freeze	During process planning, before production start

CHAPTER 40

DFMEA for CLS

Oil Pump | Pick-Up Pipe | Filter | Sensor | HEX | Coolant Pump

Item	Failure Mode	Effect	Cause	Current Controls	Recommended Action
Oil Pump	No flow output	Bearing seizure, gear damage	Pump coupling failure, air ingress	Oil pressure sensor, VCU DTC	Add redundant EOP; thermal shut
Pick-Up Pipe	Air ingestion	Pump cavitation, loss of oil	Low oil level, extreme vehicle	Suit shafting surge	Monitor under max tilt angle on c
Oil Filter	Complete blockage	Pump starvation if bypass	Filter clogging, oil bypass	Bypass valve, engine service valve	Service scheduled oil change
Winding Temp Sensor	Out-of-range reading (high)	False derating — unnecessary	Sensor operation, harness/CAN diag	Diagnostic (range check, plausibility)	Check against ambient temperature
Heat Exchanger	External fouling	Reduced heat rejection — risk	Log contamination, scale	Clean to quality specification	Monitor CSW for inlet temp
Coolant Pump	Reduced flow	Rising winding and IGBT temp	Pump wear, electrical control fault	Delta-T monitoring, SLD	Alert on abnormally high cool

CHAPTER 41

HARA — Loss of Inverter Cooling

Hazardous Event | Severity | Exposure | Controllability | ASIL

41.1 Hazard Analysis and Risk Assessment — Hazard 1

HARA Field	Detail
Hazardous Event	Loss of inverter cooling → IGBT junction overtemperature → inverter protection shutdown → unintended loss of drive
Operating Situation	Vehicle in motion on public road — overtaking manoeuvre at 80 km/h
Hazard	Sudden unintended loss of drive torque at speed on a public road
Severity (S)	S3 — Life-threatening or fatal injuries to occupants or other road users are possible if sudden torque loss causes loss
Exposure (E)	E4 — High exposure; the vehicle operates at speeds where this hazard applies for a large proportion of its operating
Controllability (C)	C2 — Most drivers can control the vehicle following sudden torque loss at 80 km/h, but some drivers in complex situ
ASIL	ASIL B (S3 × E4 × C2 per ISO 26262 Table B.1)
Safety Goal	SG1: The eDU shall not cause an unintended loss of drive torque greater than 50% of current demand within 500 ms

CHAPTER 42

HARA — Loss of Lubrication

Bearing Seizure | Axle Lock | ASIL Assignment

HARA Field	Detail
Hazardous Event	Loss of lubrication → bearing seizure → mechanical lock-up of eDU → sudden unintended loss of drive or axle lock-
Operating Situation	Vehicle travelling at 100 km/h on motorway
Hazard	Sudden unintended drive torque loss or, in worst case, partial axle lock-up causing vehicle yaw instability
Severity (S)	S3 — Axle lock-up at motorway speed could cause severe or fatal accident
Exposure (E)	E4 — Vehicle operates at motorway speeds regularly; oil system runs continuously
Controllability (C)	C1 — Sudden yaw instability from axle lock is largely uncontrollable by most drivers
ASIL	ASIL C (S3 × E4 × C1 per ISO 26262 Table B.1)
Safety Goal	SG2: The eDU shall not cause a sudden total loss of lubrication to main shaft bearings without at least 10 seconds o

CHAPTER 43

ASIL Assignment

Severity | Exposure | Controllability | Assignment Table

43.1 ASIL Parameters Explained

Parameter	Levels	Definition
Severity (S)	S0–S3	S0=no injury; S1=light injury; S2=severe/life-threatening; S3=life-threatening/fatal
Exposure (E)	E0–E4	E0=incredible; E1=very low; E2=low; E3=medium; E4=high probability during vehicle life
Controllability (C)	C0–C3	C0=controllable; C1=simply controllable; C2=normally controllable; C3=difficult to control/uncontrollable

43.2 ASIL Determination for CLS Hazards

Hazard	S	E	C	ASIL
Loss of inverter cooling → unintended torque loss	S3	E4	C2	ASIL B
Loss of lubrication → bearing seizure → axle lock	S3	E4	C1	ASIL C

CHAPTER 44

Technical Safety Requirements

Deriving TSRs from Safety Goals | DTC | Limp-Home Strategies

TSRs for Safety Goal SG1 — Loss of Inverter Cooling (ASIL B)

- TSR-SG1-001 (ASIL B): The inverter cooling system shall detect a coolant flow rate reduction to less than 50% of nominal within 1 second and set a DTC visible to the VCU. The VCU shall respond by reducing motor current to a safe thermal limit within 2 seconds of DTC assertion.
- TSR-SG1-002 (ASIL B): The inverter cold plate temperature sensor shall meet ASIL B hardware integrity requirements (dual-channel or diagnostics-compliant single channel) with a diagnostic coverage of $\geq 90\%$. A sensor out-of-range fault shall trigger a limp-home current limit without a driver warning delay exceeding 500 ms.
- TSR-SG1-003 (ASIL A): The CLS shall provide a coolant pump operational status signal to the VCU. If the pump fails to respond to a speed command within 2 seconds, the VCU shall initiate torque reduction.

TSRs for Safety Goal SG2 — Loss of Lubrication (ASIL C)

- TSR-SG2-001 (ASIL C): The oil pressure sensor on the pump outlet shall have ASIL C integrity. A drop below the minimum threshold pressure (1.5 bar at operating temperature) sustained for more than 3 seconds shall trigger a YELLOW warning lamp and reduce motor current by 30% within 500 ms.
- TSR-SG2-002 (ASIL C): If oil pressure remains below threshold for more than 10 seconds after the initial warning, the VCU shall command a controlled shutdown of the eDU (torque ramp-down to zero over 5 seconds) and illuminate a RED warning lamp, preventing restart until fault is cleared.
- TSR-SG2-003 (ASIL B): The EOP shall have a dedicated ASIL B integrity controller. Loss of EOP drive during stationary charging or low-speed operation shall trigger an immediate DTC and limp-home oil flow strategy using available mechanical pump flow.

CHAPTER 45

Requirements Cascade from Safety

Safety Goal → FSR → TSR → Hardware Requirement → OBD → Limp-Home

Safety Requirements Cascade — Full Chain

Safety Goal (ISO 26262 HARA output):

'The eDU shall not cause unintended torque loss > 50% within 500 ms without warning'

↓

Functional Safety Requirement (FSR):

'The inverter thermal monitoring function shall detect a cooling failure within 1 second'

↓

Technical Safety Requirement (TSR):

'The inverter cold plate temperature sensor shall meet ASIL B with ≥ 90% diagnostic coverage and report out-of-range within 100 ms'

↓

Hardware Safety Requirement (HSR):

'The NTC temperature sensor on the inverter cold plate shall have measurement range -40°C to +150°C, accuracy ±2°C, response time < 100 ms, connected to VCU via dedicated ASIL B-compliant signal path with short-circuit-to-ground detection'

↓

OBD / DTC Allocation:

DTC P1A23 – Inverter Cold Plate Over-Temperature

DTC P1A24 – Inverter Coolant Sensor Out of Range

Both stored in non-volatile memory; reported via UDS service 0x19

↓

Limp-Home Strategy:

On assertion of P1A23, VCU shall cap motor current to 40% rated value

Limit maintained until DTC cleared by an authorised diagnostic session

Note: This cascade must be fully documented and traceable in the requirements management tool (e.g. DOORS, Jama, Polarion) with parent-child links from Safety Goal through to hardware specification and V&V; evidence.

PART IX

Advanced Thermal Engineering

LPTN | Derating Calibration | Oil Lifetime | Winding & Magnet Temperature

CHAPTER A

Two-Loop Architecture — Oil & Coolant Circuits

Fluid Properties | Operating Ranges | Key Parameters

The two primary thermal loops of the eDU operate concurrently and are thermally coupled through the oil-to-water heat exchanger (OWC). Understanding their individual fluid properties and operating envelopes is foundational to CLS engineering.

Oil Circuit — Direct Cooling

The oil circuit uses ATF or a dedicated e-motor oil (e.g. 0W-20 synthetic). This fluid makes direct physical contact with the motor's rotating components, enabling it to remove heat from surfaces that coolant can never reach: end-winding spray cooling, rotor shaft feed, bearing lubrication, gear mesh lubrication, and differential lubrication.

Coolant Circuit — Indirect Cooling

The coolant circuit uses a water-glycol mixture (50/50 by volume) and is completely isolated from rotating and live electrical components. It removes heat from the stator housing jacket and the inverter cold plate, then rejects that heat to the vehicle radiator or chiller via the Vehicle Thermal Management System (VTMS).

Parameter	Oil Circuit	Coolant Circuit
Fluid	ATF / EV Oil (0W-20)	Water-Glycol 50/50
Density	~870 kg/m³	~1060 kg/m³
Specific Heat (Cp)	~2000 J/kg·K	~3600 J/kg·K
Typical Flow	2–10 L/min	10–25 L/min
Temp Range	–40°C to 130°C	–40°C to 115°C
Pump Type	Electric Oil Pump (EOP)	PWM Electric Water Pump

Parameter	Oil Circuit	Coolant Circuit
Heat Sink	Oil-to-Water HEX → Coolant	Vehicle Radiator / Chiller

CHAPTER B

Oil Pump Sizing — Flow Rate & Pump Speed

Heat Rejection Drives Flow | Speed-vs-Flow Map | Pump Arbitration

B.1 Core Relationship — Heat Rejection Drives Flow

A fundamental principle in EV thermal engineering is that oil flow demand is driven primarily by the heat rejection requirement, not by motor speed alone. This is in contrast to an ICE transmission where oil flow is often mechanically linked to engine speed. The electric oil pump (EOP) in an EV is fully decoupled, allowing independent thermal and lubrication control across all operating conditions.

Oil Flow Requirement Formula

$Q_{\text{required}} = P_{\text{loss}} / (\rho \times C_p \times \Delta T_{\text{allowable}})$

Q = volumetric flow rate (m³/s or L/min)
P = heat generated in component (W)
rho = oil density (~870 kg/m³ for ATF)
Cp = specific heat capacity (~2000 J/kg·K)
dT = allowable temperature rise across component (K)

B.2 Speed-vs-Flow Decision Map

Operating Condition	Speed Range	Primary Heat Source	Oil Flow Need
Low speed / high torque (launch, hill start)	0–2000 RPM	High I²R copper losses, bearing load	HIGH — maximum flow required
Mid-range cruise	3000–6000 RPM	Mixed copper + iron losses	Medium flow
High speed, light load	8000–15000 RPM	Windage losses, churning dominant	Moderate (churning losses rise sharply)
Peak / launch event	Any speed	Maximum total losses everywhere	MAXIMUM — pump at ceiling speed
Stationary with torque (hill hold)	0 RPM	Sustained I²R, zero windage	HIGH — no centrifugal oil distribution aid

B.3 Pump Speed Scheduling & Arbitration Logic

EOP Speed Arbitration — ECU Pseudocode

```
// EOP speed request arbitration (runs every 10 ms in thermal controller)
float torque_speed_req = lookup_2D(motor_torque_Nm, motor_speed_RPM);
float thermal_req = lookup_1D(MAX(T_winding_est, T_oil_sump));
float vsc_min_req = CAN_receive(VSC_MIN_FLOW_RPM_signal);

// Final command = highest of all requests (MAX wins)
float eop_target_rpm = MAX(torque_speed_req, thermal_req, vsc_min_req);

// Apply rate limit (prevents pump hunting)
eop_target_rpm = rate_limit(eop_target_rpm, prev_rpm, MAX_RAMP_UP, MAX_RAMP_DOWN);

// Convert RPM to flow using pump curve interpolation
float actual_flow = interp_2D(pump_curve_table, eop_target_rpm, system_back_pressure_bar);
```

Note: The pump H-Q curve (pressure head vs volumetric flow at each RPM) is the critical calibration input. Without an accurate H-Q curve, all flow estimates in the ECU are wrong. This curve must be measured on a flow bench at multiple oil temperatures because viscosity strongly affects pump delivery.

CHAPTER C

Electric Motor Thermal Model — LPTN

Lumped Parameter Thermal Network | Key Nodes | Stator & Rotor Models

C.1 LPTN Theory

The motor thermal model is the most important analytical tool in EV cooling engineering. It models the motor as a network of thermal resistances (R) and thermal capacitances (C) — an exact analogy to an electrical RC circuit where heat flow replaces current and temperature replaces voltage.

LPTN Governing Equations

Heat Flow (W) = Temperature Difference (K) / Thermal Resistance (K/W)
[analogous to $I = V/R$]

Thermal Capacitance (J/K) = Mass (kg) × Specific Heat Cp (J/kg·K)
[stores heat energy, causes thermal lag]

Time Constant: $\tau = R \times C$ (seconds)

C.2 Key Thermal Nodes

Node	Physical Location	Primary Heat Source	Typical Time Constant
T_copper (hot spot)	Deepest winding in slot	I²R copper loss	30–60 s
T_end_winding	Winding overhang outside slots	I²R copper loss	30–90 s

Node	Physical Location	Primary Heat Source	Typical Time Constant
T_stator_tooth	Iron core between slots	Iron eddy + hysteresis loss	60–180 s
T_stator_yoke	Outer ring of stator iron	Iron loss (lower)	120–300 s
T_housing	Stator outer cylindrical surface	Conducted from stator	300–600 s
T_magnet	Embedded in rotor	Magnet eddy currents, conducted	100–300 s
T_rotor_iron	Rotor lamination stack	Rotor iron losses	100–400 s
T_shaft	Motor shaft	Conducted from rotor	200–500 s
T_bearing	Bearing inner/outer race	Conducted + friction loss	150–400 s

C.3 Stator Thermal Model

The stator is the primary heat source in an IPM motor at most operating conditions. The slot liner — a thin sheet of electrical insulation between the copper and the stator iron — is the dominant thermal resistance in the stator heat path. Its thermal conductivity is typically only 0.2 W/m·K (compared to copper at 400 W/m·K). This is why end-winding oil spray is so thermally effective — it bypasses the slot liner resistance entirely.

Stator Loss Equations

Copper Loss = $I_{rms}^2 \times R_{dc} \times K_{ac} \times (1 + \alpha \times \Delta T)$
 K_{ac} = AC resistance factor (skin effect, proximity effect)
 α_{copper} = 0.00393 per °C

Iron Loss = $k_h \times f \times B_{peak}^2 + k_e \times f^2 \times B_{peak}^2$
 k_h = hysteresis coefficient (material-dependent)
 k_e = eddy current coefficient
 f = electrical frequency = (RPM/60) × pole_pairs
 B = peak flux density (Tesla)

C.4 Rotor Thermal Model

The rotor presents a fundamentally harder thermal problem than the stator. It rotates at up to 20,000 RPM, making direct coolant access impossible. Heat must conduct outward through solid material from magnet to rotor iron to shaft to bearing to housing — a thermal path with significantly higher resistance than the stator path. The rotor thermal time constant is 3–5 times longer than the stator, meaning rotor temperature continues rising even after electrical load is removed.

Rotor Heat Source	Speed Dependence	Torque Dependence	Peak Risk Condition
Magnet eddy currents	Strong ($\propto f^2$)	Moderate (via current harmonics)	High speed, field weakening
Rotor iron eddy loss	Strong ($\propto f^2$)	Low	High speed, light torque cruise
Rotor iron hysteresis	Linear ($\propto f$)	Low	High speed operation
Conducted from stator via air gap	Moderate (convection ↑ with Speed)	Strong (stator hotter at high torque)	Low speed, high torque

CHAPTER D

Continuous vs Peak Performance

Power-Duration Curve | Temperature Limits | Derating Algorithm

D.1 The Power-Duration Curve

Every electric motor has two distinct power ratings: a peak rating valid for a limited duration (typically 10–30 seconds), and a continuous rating that can be sustained indefinitely. The boundary between these regions is entirely defined by the thermal management system. A better cooling system directly translates into a higher continuous rating and a longer peak duration.

Peak Duration Calculation

Continuous Power: operating point where $dT_{winding}/dt = 0$
(thermal equilibrium reached with $T_{winding} < T_{limit}$)
Cooling system MUST reject all losses at steady state.

Peak Duration:
 $t_{peak} = (m \times Cp \times \Delta T_{allowed}) / P_{loss_excess}$
 m = thermal mass of motor (kg)
 Cp = weighted specific heat (J/kg·K)
 $\Delta T = T_{limit} - T_{initial}$
 P_{loss} = heat generation at peak operating point (W)

D.2 Temperature Limits by Component

Component	Continuous Limit	Peak Limit	Failure Mode if Exceeded
Stator winding (Class H)	180°C	200–210°C	Insulation breakdown → short circuit
Stator winding (Class C)	200°C+	220°C+	Insulation carbonisation
Rotor magnet (NdFeB N45SH)	120°C	140–150°C	Irreversible demagnetisation
Bearing (all types)	120°C	150°C	Fatigue spalling, seizure
Oil sump	125°C	135°C	Oil oxidation, varnish, seal failure
IGBT junction (Si)	150°C	165°C	Thermal runaway, device failure
SiC MOSFET junction	175°C	185°C	Bond wire failure, short
Gear steel surface	120°C	140°C	Scuffing, micropitting

D.3 Thermal Derating Algorithm

Multi-Input Torque Limit Arbitration


```
// Multi-input torque limit arbitration in motor controller:
Torque_limit = MIN(
f(T_winding), // winding insulation protection
f(T_magnet), // demagnetisation protection
f(T_junction), // inverter IGBT/SiC protection
f(I_phase_limit), // current limit (temp-dependent)
f(V_dc, RPM), // voltage/flux limit
f(T_bearing), // bearing protection
f(T_oil_sump) // oil degradation protection
)
```

CHAPTER E

Drive Cycles & Testing — Component to Vehicle

Motor Tests | EDU Integration | Vehicle Validation | Proving Ground

E.1 Motor Component Level Tests — Thermal Mapping

The motor thermal mapping test is the foundational dataset for the entire thermal engineering programme. It measures steady-state temperatures at every point in the motor's operating envelope and is used to calibrate the LPTN thermal model.

Thermal Mapping Test Procedure

Test Setup:

Motor on dyno with controlled external coolant and oil supplies

Thermocouples: end winding (×4), stator tooth (×2), yoke (×2), housing (×4),
oil in/out, coolant in/out, bearings (×2)

Flow meters on both oil and coolant circuits

Test Matrix:

Torque: 50, 100, 150, 200, 250, 300 Nm

Speed: 500, 1000, 2000, 4000, 8000, 12000 RPM

T_coolant_in: 40, 65°C

T_oil_in: 60, 90°C

Oil flow: 3, 6, 9 L/min

Coolant flow: 10, 15, 20 L/min

Procedure per point:

1. Start from cold soak (all temps = 25°C)
2. Apply constant torque and speed
3. Wait for equilibrium: $dT/dt < 0.5^\circ\text{C}/\text{min}$ on all sensors
4. Record all temperatures, flows, pressures
5. Cool down fully before next point

Total points: 200–400 steady-state operating points

Typical duration: 3–6 weeks of dyno time

E.2 EDU Integration Tests — Drive Cycle Simulation

At EDU level, motor, inverter, and gearbox are tested as an integrated assembly. New thermal phenomena emerge at this level: inverter switching losses heat the coolant before it enters the motor; gearbox churning losses heat the oil before it reaches the end windings; real oil distribution across multiple jets can be measured.

E.3 Worst-Case Thermal Stress Tests

Test Name	Procedure	What It Validates
Repeated Launch	Full peak torque 0→100 km/h, coast back, repeat until demand is met	Peak thermal capacity, derating activation
Sustained Grade Climb	Simulate 10% grade at 120 km/h for 30 minutes continuously	Continuous rating at worst sustained condition
Hot Ambient Grade	As above but with coolant inlet forced to 70°C	True worst-case combined condition
Hot Soak Restart	Motor at 80°C thermal soak, immediately apply peak torque demand	Derating in hot restart — customer worst case
Cold Start High Load	–20°C cold soak, full torque demand immediately	Oil viscosity, pump priming, cold start protection

E.4 Vehicle Level Validation — Proving Ground Tests

Location	Key Tests	Pass Criteria
Arizona (Death Valley, July)	Sustained motorway cruise, towing in hills, repeated hard acceleration	No derating WLTP; continuous grade sustained without torque reduction
Sweden / Northern Canada (January)	Cold start oil viscosity validation, pump priming, warm-up	Pump, heater demands, interface within 2 min
Alps / Pikes Peak	Sustained grade climb; repeated ascent for thermal soak	Continuous thermal scenario managed; no overtemperature
Dubai / Abu Dhabi	45°C + high humidity — dual stress on cooling and sealing	All temperature limits maintained; no seal failure

CHAPTER F

VSC Minimum Oil Flow Request

System Flow Floor | Post-Drive Strategy | Software Implementation

F.1 What Is a VSC Request?

The Vehicle System Controller (VSC) — sometimes called the VCU — is the top-level vehicle brain that coordinates all subsystems. It sends flow requests to the thermal management system, including the electric oil pump, based on system-level needs that go beyond what the motor torque and speed demand would require. The VSC minimum flow request acts as a hard floor below which the oil pump speed never falls.

F.2 Reasons for Minimum Flow

Reason	Explanation	Consequence If Not Met
Bearing lubrication	Bearings need minimum oil film even at standstill	Bearing surface damage, increased wear rate
Seal conditioning	Lip seals and O-rings require wetted contact to prevent leakage	Oil leakage, seal replacement, leakage path formation
Thermal soak protection	After hard driving, motor is thermally hot. Oil must keep circulating to remove residual heat	Over-temperature, potential demag risk for magnets
Gear pre-lubrication	Before first torque application, gear and differential surfaces need oil film	Initial surface wear, scuffing
Thermal model boundary	Thermal observer in ECU needs stable oil inlet temperature to model heat transfer	Model prediction error, next torque application
Post-drive run-on	Driver switches off; pump continues running until T_winding < 60°C	Islanding, core cooled unnecessarily; demag risk for magnets

F.3 Typical VSC Minimum Flow Values

Operating Condition	Min Oil Flow	Approx Pump Speed	Duration
Ignition ON, cold, stationary	0.5–1.5 L/min	200–500 RPM	Until first drive
Hot soak (T_winding > 80°C), vehicle stopped	1.5–3.0 L/min	500–1000 RPM	Until T_winding < 60°C
Post-drive run-on (ignition off)	1.0–2.0 L/min	400–800 RPM	2–10 minutes
Active driving, zero torque coasting	1.0–2.0 L/min	400–800 RPM	Continuous while moving
Remote pre-conditioning (parked)	1.0 L/min	350–600 RPM	Duration of preconditioning
Full performance demand	6–10 L/min	2000–4000 RPM	Duration of demand

CHAPTER G

Requirements Cascade — Vehicle to Motor Level

EDU Requirements | Motor Requirements | Thermal Budget Allocation

G.1 EDU Level Requirements

EDU Requirement	Value	Derived From
Peak output torque	456 Nm	0–100 km/h in 5.5 s
Peak output power	~250 kW	Top speed + grade target
Continuous output torque	180 Nm	20% grade at 80 km/h sustained
Maximum output speed	8,500 RPM	200 km/h top speed
Peak duration	≥ 30 seconds	Product requirement — launch feel
EDU efficiency (WLTP average)	≥ 93%	500 km range target

EDU Requirement	Value	Derived From
Coolant inlet temperature max	70°C	System thermal budget
Oil sump temperature max	130°C	Sealed-for-life oil requirement
No derating in WLTP	Full cycle, any ambient to 40°C	Homologation requirement

G.2 Motor Level Requirements

Motor Requirement	Value	Derived From
Peak torque	350 Nm (≤ 10 s)	EDU peak torque $\div$ gear ratio
Continuous torque	160 Nm (steady state)	EDU continuous $\div$ gear ratio
Peak power	230 kW	EDU peak power – inverter losses
Maximum speed	16,000 RPM	EDU max speed $\times$ gear ratio
Base speed (corner point)	3,500 RPM	Motor design — MTPA/FW boundary
Winding temperature limit	180°C continuous / 200°C peak	Insulation class selection (Class H)
Magnet temperature limit	120°C continuous / 140°C peak	Magnet grade N45SH properties
Coolant inlet temp max	70°C	EDU level requirement
Oil inlet temp max	90°C	Sump + HEX thermal budget
Minimum coolant flow	10 L/min	Heat rejection capacity calc
Minimum oil flow	3 L/min	End winding temp at continuous

G.3 Thermal Budget Allocation

Thermal Budget & Flow Requirement Derivation

Total heat rejection budget:
 $P_{loss_total} = P_{input} \times (1 - \eta_{EDU}) = 250 \text{ kW} \times (1 - 0.93) = 17,500 \text{ W}$

Loss allocation:
Motor copper loss ~6,000 W (34%)
Motor iron loss ~2,500 W (14%)
Inverter switching loss ~3,500 W (20%)
Inverter conduction ~2,000 W (11%)
Gearbox mesh loss ~2,000 W (11%)
Bearing + windage ~1,500 W (9%)
Total: 17,500 W

Coolant circuit rejection (inverter + stator jacket):
 $P_{coolant} = 3,500 + 2,000 + 6,000 \times 0.4 = \sim 8,000 \text{ W}$
 $Q_{coolant} = 8,000 / (1060 \times 3600 \times 10) = 12.6 \text{ L/min} \rightarrow \text{spec: } 13 \text{ L/min}$

Oil circuit rejection (end winding + rotor + gear + bearing):
 $P_{oil} = 6,000 \times 0.6 + 2,500 + 2,000 + 1,500 = \sim 9,600 \text{ W}$
 $Q_{oil} = 9,600 / (870 \times 2000 \times 15) = 3.7 \text{ L/min} \rightarrow \text{spec: } 4 \text{ L/min minimum}$

CHAPTER H

Derating Calibration

Thresholds | Two-Zone Map | Rate Limiting | Hysteresis

H.1 Why Derating Calibration is Critical

Calibration Error Direction	Consequence
Too aggressive (derate too early)	Performance unnecessarily restricted; customer complaint 'car feels slow when warm'; competitor disadvantage
Too conservative (derate too late)	Winding temperature exceeds insulation class limit; accelerated ageing; possible catastrophic failure

H.2 Two-Zone Derate Map

Two-Zone Derating Map Structure

```

T_winding < T_derate_start:
Torque_limit = 100% (no restriction)

T_derate_start <= T <= T_derate_end:
Linear reduction:
Torque_limit(%) = 100 × (T_derate_end - T_winding) / (T_derate_end - T_derate_start)

T_winding > T_derate_end:
Torque_limit = 0% (full derate – full protection)

Typical values (Class H motor):
T_derate_start = 160°C
T_derate_end = 175°C
T_recovery = 145°C (hysteresis – prevents hunting)

```

H.3 Setting T_derate_end

T_derate_end Calibration

```

T_derate_end = T_insulation_class - Safety_margin

Safety margin sources:
Model prediction error: ±5 to 8°C → margin needed
Sensor/input uncertainty: ±2 to 3°C → margin needed
Aging factor (end of life): insulation degrades → extra margin
Regulatory safety factor: programme-specific

Typical result for Class H motor:
T_derate_end = 170 to 175°C (180°C rated minus 5–10°C model uncertainty margin)

Magnet derate end (separate limit):
T_demag (N45SH at zero current) = 180°C
T_demag (N45SH at peak FW current) = 120°C
T_derate_end_magnet = 130 to 140°C

```

H.4 Rate Limiting & Hysteresis

Rate Limiting & Hysteresis Calibration

Rate Limiting:
Maximum derate rate: -50 Nm/s (negative = reducing)
From 350 Nm to 0 Nm: minimum 7 seconds
Maximum recovery rate: +20 Nm/s (gentler than derate)

Verification:
 $dT/dt \times t_{\text{ramp_down}} < \text{delta_T_available}$
Example: $0.53^{\circ}\text{C/s} \times 7\text{s} = 3.7^{\circ}\text{C}$ rise during ramp
 $3.7 < (175 - 160) = 15^{\circ}\text{C} \rightarrow$ safe with margin

Hysteresis:
Derate activates at: $T_{\text{derate_start}} = 160^{\circ}\text{C}$
Derate fully removed: $T_{\text{recovery}} = 145^{\circ}\text{C}$
Gap = 15°C prevents hunting between active/inactive states

WITHOUT hysteresis:
Temperature oscillates either side of $T_{\text{derate_start}}$
 $\rightarrow$ repeated torque steps $\rightarrow$ driveability issue

CHAPTER I

Oil Lifetime Profile

Degradation Mechanisms | Arrhenius Analysis | Oil Selection Criteria

I.1 Why EV Oil Lifetime Is Uniquely Challenging

EV powertrain oil faces a unique combination of stresses not encountered in conventional ICE transmissions. Most significantly, many modern EV EDU designs are sealed-for-life — meaning the factory-filled oil must survive the entire vehicle service life of 200,000 km and 15+ years with no oil change. This makes the oil lifetime analysis a direct input to the cooling system design requirement.

Stress Factor	ICE Transmission	EV Powertrain
Electrical stress	None (no live conductors in oil)	HIGH — copper windings bathed in oil
Operating speed	Moderate (<6,000 RPM gears)	VERY HIGH (shaft to 20,000 RPM)
Thermal load	Moderate, cyclic	High, sustained at peak performance
Service interval	60,000–100,000 km	Sealed for life (no drain plug in many designs)
Contamination	Combustion products (soot, fuel)	Copper ions, insulation particles, wear metals
Electrical conductivity	Not a concern	Critical — must remain low throughout life

I.2 The Five Degradation Mechanisms

1 — Thermal Oxidation (Primary)

Thermal oxidation is the dominant oil degradation mode. At elevated temperatures, oil molecules react with dissolved oxygen in a chain reaction that produces acidic byproducts, varnish, and sludge. The Arrhenius relationship means that every 10°C increase in oil temperature approximately doubles the oxidation rate — making sump temperature control a direct life-of-oil requirement.

Arrhenius Oxidation Rate
Rate doubles every ~10°C rise: At 100°C: oil life = L (reference) At 110°C: oil life = L/2 At 120°C: oil life = L/4 At 130°C: oil life = L/8 At 140°C: oil life = L/16

2 — Electrical Degradation (EV-Specific)

Mechanism	Description	Consequence
Electrostatic discharge erosion	PWM inverter creates shaft voltage → bearing currents discharging through oil	Bearing discharge through oil particles in oil
Copper ion dissolution	Copper dissolves from winding surfaces into oil; oxidation rate accelerated at 10x	Oil oxidation rate accelerated 2-10x even at 5–10 ppm Cu
Conductivity increase	Degraded oil becomes more electrically conductive	Current leakage paths; insulation risk
Additive electrolysis	Polar additive molecules migrate under electric field	Add-wear additives depleted at critical surfaces

3 — Mechanical Shear Degradation

High-speed gear mesh and bearing contacts mechanically shear long-chain polymer additives (viscosity index improvers) in the oil, causing permanent viscosity loss. As viscosity drops, the elastohydrodynamic (EHD) oil film thickness at gear teeth and bearing contacts reduces, increasing metal-to-metal contact and wear rate.

4 — Hydrolysis (Water Contamination)

Water enters the oil circuit through condensation during thermal cycling, seal permeation over time, and residual manufacturing moisture. Water reacts with additives causing hydrolysis, rust, emulsification, and — most critically for EVs — forms copper hydroxide with dissolved copper ions, which strongly accelerates oxidation. Water content must be kept below 200 ppm.

5 — Particle Contamination

Particle Size	Effect	Source
>10 µm	Direct bearing damage — three-body abrasive wear	Gear/bearing wear debris, assembly contamination
1–10 µm	Accelerated surface fatigue — micropitting on gear teeth	Seal wear, oxidation products
<1 µm	Catalytic acceleration of oxidation	Insulation particles, very fine wear debris

1.3 Arrhenius Lifetime Calculation

Cumulative Damage Model — Miner's Rule
--

The stator winding hot spot is the single most important temperature in the entire EV powertrain thermal system. It is the temperature at the innermost copper conductor in the deepest slot — the furthest point from any cooling surface. It is never directly measured in production vehicles; the ECU thermal observer estimates it from measurable boundary conditions.

Slot Thermal Resistance Chain

Slot thermal conductivities:

Copper: 400 W/m·K (excellent)

Epoxy impregnation: 0.2–1.0 W/m·K (moderate – fills voids)

Slot liner (Nomex/Kapton): 0.15–0.25 W/m·K ← BOTTLENECK

Silicon steel: 25–40 W/m·K (good)

Aluminium housing: 150–200 W/m·K (good)

Heat path: [Copper] → R_enamel → [Slot varnish] → R_slot_liner
→ [Stator tooth iron] → R_tooth → [Stator yoke]
→ R_housing_coolant → [Coolant]

Slot liner is the dominant resistance – even 0.2 mm at 0.2 W/m·K creates significant delta_T. This is why end-winding oil spray is so thermally valuable: it bypasses the slot liner resistance entirely.

J.2 Insulation System Components

Layer	Material	Thickness	Function	Failure Mode
Conductor enamel	Polyimide (PI) or PAI	30–80 µm	Isolate conductors from each other	Turn-to-turn short circuit
Slot liner	Nomex or Kapton film	0.2–0.5 mm	Isolate winding from stator iron (ground)	Winding-to-ground fault
Phase separator	Nomex paper	0.2–0.3 mm	Separate phase conductors in shared slots	Phase-to-phase arc fault
Impregnation varnish	Epoxy or polyester resin	Fills all voids	Mechanical bonding + thermal conductivity	Cracking, moisture ingress → insulation loss

J.3 Thermal Ageing — The Montsinger Rule

Insulation does not fail instantly when its class temperature is exceeded. It undergoes cumulative thermal ageing — an irreversible chemical degradation that progressively reduces dielectric strength and mechanical properties.

Montsinger Rule — Insulation Life vs Temperature

Montsinger Rule: 'Every 10°C rise in operating temperature halves insulation life'

Arrhenius form:

$L(T) = L_{ref} \times \exp(Ea/R \times (1/T - 1/T_{ref}))$

Ea = activation energy (~1.0–1.4 eV for polyimide)

T = operating temperature (Kelvin)

Class H example (L_ref = 20,000 hrs at 180°C):

At 170°C → life = 40,000 hrs (2× longer)

At 160°C → life = 80,000 hrs (4× longer)

At 190°C → life = 10,000 hrs (2× shorter)

At 200°C → life = 5,000 hrs (4× shorter)

J.4 Key Winding Temperature Thresholds

Temperature	Significance
20°C	Cold soak reference — calibration baseline; maximum copper resistance accuracy
60°C	Oil typically reaches this temperature before motor on warm-up; VSC may allow full torque
100°C	Copper resistance 31% above cold; efficiency measurably lower; normal operating range
130°C	Typical steady-state at aggressive but normal driving; continuous operation target
155°C	Class F insulation rated limit; derate warning zone in older motor designs
160°C	Typical T_derate_start for Class H motor; controlled torque reduction begins
175°C	Typical T_derate_end; full torque removal; model uncertainty margin before class limit
180°C	Class H continuous rated limit; NOT to be exceeded in normal operation under any condition
200°C	Short-term absolute maximum; insulation survives briefly; life significantly consumed
220°C+	Irreversible insulation damage; slot liner breakdown risk; immediate hardware failure risk

J.5 Measurement Methods in Development

Method	What It Measures	Advantages	Limitations
Embedded thermocouples (Type K)	Temperature inside slot and at end winding	Direct measurement; fast response	Spatial coverage limited; multiple sensors needed for full slot
Resistance Temperature (DC method)	Average copper temperature from winding voltage drop	Winding voltage drop temperature; non-invasive	Average only; cannot detect hot spot; not real-time; stops at 180°C
Infrared thermal camera	End winding surface temperature distribution	Spatial temperature map; identifies hot spots	Surfaces only; uneven emissivity; requires calibration
Fibre optic sensor	Point temperature inside slot with thin probe	High accuracy; immune to electrical noise	Fragile; expensive; requires special slot preparation

CHAPTER K

Rotor & Magnet Temperature

Demagnetisation Risk | Cooling Strategies | Stator-Rotor Coupling

K.1 Why the Rotor Is the Harder Thermal Problem

Despite often generating less total heat than the stator, the rotor presents a fundamentally harder thermal problem for three reasons: it rotates at up to 20,000 RPM making direct coolant access impossible; the thermal resistance from magnet to coolant is 5–10 times higher than from stator winding to coolant; and the long thermal time constant means rotor temperature can continue rising for minutes after electrical load is removed — the thermal soak phenomenon.

Rotor vs Stator Thermal Comparison

Stator thermal resistance (winding to coolant): ~5-15 K/kW
Rotor thermal resistance (magnet to coolant): ~30-80 K/kW

Rotor thermal time constant: 100-500 seconds
Stator thermal time constant: 30-120 seconds

Result: Rotor temperature rises SLOWLY but KEEPS RISING even after electrical load is removed. Post-drive thermal soak is critical.

K.2 Magnet Properties vs Temperature

NdFeB (Neodymium Iron Boron) permanent magnets are used in virtually all high-performance EV traction motors. Their magnetic properties — remanence Br and coercivity Hci — both degrade with temperature. Critically, coercivity is far more temperature-sensitive than remanence.

Temperature	Br (T)	Hci (kA/m)	Relative Demagnetisation Risk
20°C	1.35	1600	Very Low — full properties
60°C	1.31	1380	Low
80°C	1.28	1270	Low-Moderate — normal warm operation
100°C	1.26	1150	Moderate — field weakening care needed
120°C	1.23	1020	High — continuous limit
140°C	1.21	900	Very High — peak limit
160°C	1.18	780	CRITICAL — avoid under all conditions

K.3 Demagnetisation — The Irreversible Failure Mode

Demagnetisation is the most feared failure in EV motor design. Unlike insulation degradation which is gradual and potentially detectable, demagnetisation can be sudden and leaves the motor permanently damaged with reduced torque capability that cannot be restored by any repair short of magnet replacement.

Demagnetisation Conditions

Condition 1 – High Temperature Alone:
Hci decreases → knee point of B-H curve rises
Even without high current, magnet can demagnetise at elevated temperature
Defines: T_demag_limit (material property of magnet grade)

Condition 2 – High Current + Elevated Temperature:
During field weakening, large negative d-axis current creates demagnetising field
At elevated temperature (low Hci), this field can exceed knee point
→ partial or full demagnetisation

Most critical during:

- High speed operation (field weakening region)
- Short circuit fault current event
- Peak torque demand when rotor already hot

Consequence of partial demagnetisation:
5-20% permanent torque reduction – cannot be repaired – motor replacement required

K.4 Rotor Cooling Strategies

Strategy	Description	Thermal Effectiveness	Design Challenge
Shaft oil feed	Oil pumped through hollow shaft; centrifugal forces distribute radially	High if flow distribution is adequate	Sealing at input; uniform distribution; high-speed re
End plate oil spray	Oil jets directed at rotor end faces; cools stator end windings and proximal	MODERATE and targeted	Containing high-speed flywheels away at high RPM; coverage
Air gap oil mist	Forced oil mist through air gap; direct rotor surface contact	HIGH – direct contact	Major windage loss increase; foam generation; air gap m
Passive conduction via stator	Rotor heat → air gap convection → stator cooling	LOW – high air gap resistance	Background path only — not a primary cooling strategy

K.5 Operating Condition vs Thermal Risk Summary

Operating Condition	Primary Thermal Risk	Reason	Cooling Priority
Low speed, high torque (launch, hill hold)	STATOR WINDING	High I²R; low air gap convection; no centrifugal distribution	Maximum oil flow; stator winding spray critical
High speed, light load (motorway cruise)	ROTOR MAGNET	High iron losses at high electrical frequency	Shaft oil feed; magnet cooling spray
High speed, high torque (performance)	BOTH simultaneously	Maximum losses everywhere; worst-case combined condition	Maximum combined cooling
Regenerative braking	Usually LOW — monitor	Current flows but energy recovered; stator I²R	Stator I²R maintained; no extra demand
Stationary with torque (hill hold)	STATOR — worst case	Zero speed; zero windage; sustained current	Maximum oil flow; ifugate cooling; time-limit if needed
Post-drive thermal soak	ROTOR MAGNET	Long rotor time constant; temperature persists after load pumping	Essential

CHAPTER

Quick Reference Tables

Key Numbers | Limits | Flow Rates | Material Properties

Thermal Limits Quick Reference

Component	Continuous Limit	Peak Limit	Hard Absolute Limit
Stator winding (Class F)	155°C	170°C	180°C
Stator winding (Class H)	180°C	200°C	220°C
Rotor magnet (N45SH)	120°C	140°C	150°C
Bearing (standard)	120°C	140°C	160°C
Oil sump	125°C	135°C	145°C
IGBT junction (Si)	150°C	165°C	175°C
SiC MOSFET junction	175°C	185°C	200°C

Flow Rate Reference

Circuit / Condition	Min Flow	Nominal Flow	Max Flow
Oil — VSC minimum (standstill)	0.5–1.5 L/min	—	—
Oil — continuous operation	2–3 L/min	4–6 L/min	8–10 L/min
Oil — peak performance	6–8 L/min	8–10 L/min	10–12 L/min
Coolant — stator jacket	5–8 L/min	10–15 L/min	18–22 L/min
Coolant — inverter cold plate	5–8 L/min	8–12 L/min	15–18 L/min
Coolant — combined EDU	10–12 L/min	15–20 L/min	22–28 L/min

Key Material Properties

Material	Density (kg/m³)	Cp (J/kg·K)	Thermal Cond. (W/m·K)
Copper (winding)	8,900	385	400
Silicon steel (lamination)	7,600	460	25–40
Aluminium (housing)	2,700	900	150–200

Material	Density (kg/m³)	Cp (J/kg·K)	Thermal Cond. (W/m·K)
NdFeB magnet	7,500	420	6–9
ATF/EV oil (100°C)	~830	~2100	~0.14
Water-glycol 50/50 (70°C)	~1040	~3700	~0.42
Slot liner (Nomex/Kapton)	~1,400	~1,100	0.15–0.25
Epoxy impregnation	~1,200	~1,000	0.2–1.0

Copper Resistance Temperature Correction

Winding Temperature	R/R_20 Factor	Copper Loss Multiplier vs Cold
20°C	1.000	1.00× (reference)
60°C	1.157	1.16× — warm motor
100°C	1.314	1.31× — normal operating
130°C	1.431	1.43× — aggressive driving
160°C	1.549	1.55× — derate zone
180°C	1.627	1.63× — class limit

Insulation Class Summary

Class	Max Continuous Temp	Typical Material	EV Application
Class F	155°C	Epoxy resins, glass fibre	Early EV designs; budget applications
Class H	180°C	Silicone elastomers, polyimide	Most current production EV motors
Class C	>180°C	Polyimide, PTFE, ceramic	High performance EVs; motorsport
200 / 220	200–220°C	Advanced polyimide systems	Next-gen high-performance EV motors

Key Terminology Glossary

ASIL	Automotive Safety Integrity Level — a risk classification per ISO 26262 from QM to D
BDD	Block Definition Diagram — SysML diagram defining block types and their properties
CLS	Cooling and Lubrication System — the system boundary for this learning programme
DTC	Diagnostic Trouble Code — a standardised fault identifier stored by the VCU/ECU
EARS	Easy Approach to Requirements Syntax — structured template for writing requirements
eDU	Electric Drive Unit — integrated motor, inverter, and gearbox assembly

EOP	Electric Oil Pump — independently electrically driven oil pump in the eDU
FMI/FMU	Functional Mockup Interface / Unit — open standard for model exchange in simulation
HARA	Hazard Analysis and Risk Assessment — ISO 26262 process for identifying hazardous events
HIL	Hardware-in-the-Loop — test method where physical hardware is integrated into a simulation loop
IGBT	Insulated Gate Bipolar Transistor — power semiconductor used in inverter bridges
LPTN	Lumped Parameter Thermal Network — RC-analogy thermal model of the motor
MBSE	Model-Based Systems Engineering — using formal models as primary artefacts in SE
NTC	Negative Temperature Coefficient — thermistor type used for temperature measurement
OWC	Oil-to-Water heat exchanger — transfers heat from oil circuit to coolant circuit
PFMEA	Process Failure Mode and Effects Analysis — FMEA focused on manufacturing processes
SiC	Silicon Carbide — wide bandgap semiconductor for high-frequency, high-efficiency inverters
TSR	Technical Safety Requirement — safety requirement derived from a Safety Goal for ASIL allocation
VCU	Vehicle Control Unit — top-level vehicle controller managing all powertrain subsystems
VTMS	Vehicle Thermal Management System — the vehicle-level system managing all thermal circuits
WLTP	Worldwide Harmonised Light Vehicle Test Procedure — regulatory drive cycle for EV range testing

Standards and References

- ISO 26262:2018 — Road vehicles: Functional safety
- ISO 21448:2022 — Safety of the intended functionality (SOTIF)
- SAE J2908 — Electric/hybrid powertrain test procedures
- IEC 60085 — Thermal evaluation and designation of electrical insulation
- EARS (Easy Approach to Requirements Syntax) — Mavin et al., 2009 IEEE RE Conference
- SysML v1.6 — Systems Modelling Language specification (OMG)
- IPG Automotive CarMaker Documentation — <https://www.ipg-automotive.com>
- Mechanical Simulation Corporation CarSim Documentation — <https://www.carsim.com>
- JLR Internal — eDU CLS Design Intent Document (programme-specific)
- JLR Internal — VTMS Interface Control Document (programme-specific)

*End of Document — EV Powertrain Cooling & Lubrication — Systems Engineering Reference
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